
Derivatives Clearing A Comprehensive Guide

**Risk Management, Regulation, and the
Future of the Markets**

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M A N U S P R E S S

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A Reader's Compass

Reader Profile	Key Interests	Recommended Chapters
Practitioners at financial institutions (Operations / IT)	Day-to-day workflows, system requirements	Chapters 1, 3, 6; related columns
Practitioners at financial institutions (Risk Management / Compliance)	Risk models, regulatory developments, default management	Chapters 2, 4, 5; related columns
Buy side (asset managers, funds, etc.)	Cost efficiency, broker selection, asset protection	Chapters 3, 6, 7; related columns
Students and researchers	Systematic view of the whole, historical background	Read Chapters 1–2 first, then onward; history-related columns

Disclaimer

This book is intended to provide general information on derivatives trading and on clearing systems and practices. It does not constitute a recommendation of any specific transaction, investment decision, or legal or tax judgment. While every effort has been made to ensure the accuracy, completeness, and currency of its contents, no warranty is given in this regard, and neither the author nor any related parties accept any liability for losses arising from the use of this book.

The views expressed in this book are those of the author alone and do not represent the official positions of the author's employer or of any exchange, clearing house, regulatory authority, or industry body. Systems, market practices, and regulations are subject to change; before applying the contents of this book in practice, readers should consult the latest official documents and rules and seek professional advice.

Introduction: The Evolution and Strategic Imperative of Derivatives Clearing

Overview

In contemporary financial markets, derivatives serve as an indispensable catalyst for economic activity, performing vital functions such as risk hedging, price discovery, and capital allocation. The scale of this market is profound — the notional outstanding of over-the-counter (OTC) derivatives alone exceeds USD 700 trillion (BIS, 2023). For these massive volumes to be transacted seamlessly and securely, a highly sophisticated ecosystem must function flawlessly across the sequential lifecycle of execution, clearing, and settlement.

At the vortex of this ecosystem lies the subject of this book: clearing — the pivotal mechanism that safeguards both systemic market stability and individual transactional integrity. Through clearing, a central counterparty (CCP) interposes itself between the original contracting parties, becoming the buyer to every seller and the seller to every buyer. In doing so, the CCP isolates and assumes counterparty credit risk (the risk of participant default) and guarantees the ultimate performance of the contract.

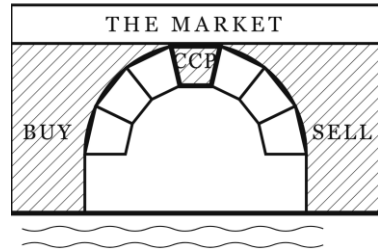
While an extensive and excellent literature exists on derivative instruments, pricing methodologies, and market risk management — most notably John Hull's seminal *Options, Futures, and Other Derivatives* (Hull, 2021) — comprehensive treatises dedicated systematically to clearing remain exceedingly scarce, particularly in the Japanese language. This book seeks to fill that critical void by providing a rigorous examination of clearing mechanics, risk management frameworks, regulatory architectures, and recent structural shifts.

My perspective is shaped by more than two decades in front-office leadership roles at global financial institutions, spanning listed futures, options, and OTC derivatives.

Having dedicated the first decade of my career to execution, electronic trading, and algorithmic frameworks, I have spent the years following the 2008 global financial crisis at the vanguard of clearing infrastructure and its evolutionary trajectory. This dual experience crystallized a fundamental truth: clearing is far from a routine administrative back-office

function. Rather, it is an intellectually demanding, strategic discipline that requires sophisticated risk modeling, a command of labyrinthine regulations, and the deployment of cutting-edge financial technology. Regrettably, within the Japanese financial landscape, clearing is often still mischaracterized as a mere operational afterthought. This book is driven by a desire to correct this perception, demonstrating the strategic density and intellectual allure of this critical domain.

DIVE INTO CLEARING



The systemic imperative of clearing escalated dramatically in the wake of the 2008 global financial crisis, which exposed the severe vulnerabilities latent in an opaque, un-cleared OTC derivatives market. The subsequent G20 mandate orchestrated a sweeping overhaul of market architecture, transforming central clearing via CCPs into a mandatory pillar for standardized contracts, alongside stringent margin requirements for non-centrally cleared transactions.

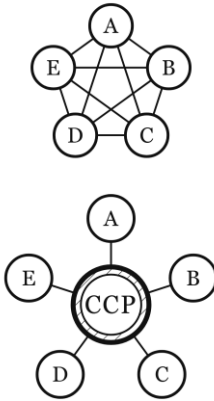
This historic regulatory pivot thrust clearing into the absolute center of the financial ecosystem. In the ensuing years, the discipline has evolved with remarkable velocity: CCP risk-management tools have grown highly sophisticated; client clearing

structures have become intricately nuanced; and regulatory frameworks have continuously adapted. Over the past fifteen years, clearing has arguably been one of the most dynamic and transformative sectors in global finance. Crucially, academic scholarship has advanced in tandem, establishing central clearing as a rigorous discipline in its own right — one that sits at the vital intersection of quantitative risk management, market microstructure, and the economic theory of financial regulation.

Purpose and Structure of This Book

This book aims to explain the ever-evolving world of derivatives clearing **broadly, systematically, and practically** — from its basic concepts to the latest developments and the outlook for its future. By unraveling the complexity and depth of clearing and conveying its strategic importance, it seeks to make the field engaging for readers encountering it for the first time, while offering practitioners and researchers already engaged in it a framework for organizing their knowledge and a source of fresh perspectives.

The book is structured so that readers can build up a complete picture of clearing step by step. The chapter-by-chapter introduction below is offered as a guide for navigating the book according to your own interests and needs.

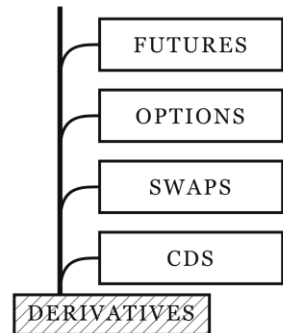


Chapter 1, “Fundamentals of Clearing”:

This chapter is the gateway to the world of clearing. It begins by locating clearing within the trading–clearing–settlement ecosystem of a derivatives transaction. It then takes up the fundamental question of why clearing is necessary, clarifying its significance for reducing counterparty risk, guaranteeing settlement, and enhancing market transparency. The basic advantages and disadvantages of clearing, the principal participants

(CCPs, clearing members, and clients), and essential terminology such as margin are explained in plain language, so that even first-time readers can build a solid foundation. It is the first step toward understanding clearing’s importance as market infrastructure.

Chapter 2, “An Introduction to Cleared Products”: To which financial instruments does clearing actually apply? This chapter focuses on the principal derivative products subject to clearing. From traditional exchange-traded derivatives such as futures and options to OTC derivatives such as interest rate swaps, credit default swaps (CDS), and FX derivatives, it explains each product’s basic mechanics, the background to its trading, and its particular risks. It also discusses why each product is cleared by a CCP — or subject to a clearing mandate — making clear the connection between product knowledge and clearing. Recommended for readers who want to organize their knowledge of derivative products or to see concretely where clearing applies.



Chapter 3. “Clearing Mechanics and Workflow”: How does the clearing process

CLEARING MECHANICS
AND WORKFLOW

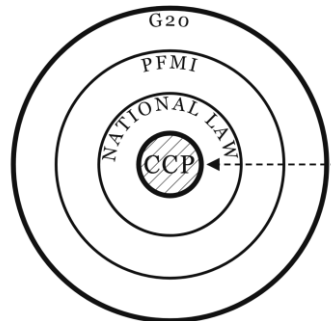


actually unfold? This chapter traces in detail the full operational lifecycle from the moment a trade is executed, through clearing at the CCP, to final settlement: how trade data is submitted to the CCP, matched, and accepted; how positions are managed and netted day to day; the

calculation (including daily mark-to-market), calling, and exchange of margin; the computation of interest and fees; and expiry processing at contract maturity, whether by cash settlement or physical delivery. It also covers the practical flow of information and system connectivity among the CCP, clearing members, and clients, making it essential for understanding the reality of clearing operations. Particularly useful for operations staff and those involved in systems development.

Chapter 4. “Legal and Regulatory Frameworks and Regional Characteristics”:

Clearing is strictly governed by the laws and regulations of each country and region. This chapter surveys the regulatory landscape: the international currents that originated in the post-2008 G20 agreements (the Basel framework, the CPMI-IOSCO Principles for Financial Market Infrastructures, and others) and the specific regimes of the major jurisdictions — the Dodd-Frank Act in the United States, EMIR in Europe, and the Financial Instruments and Exchange Act in Japan — including clearing mandates, margin requirements, and trade reporting obligations, compared side by side. It further develops a global perspective on differences in market practice across jurisdictions, divergences in how



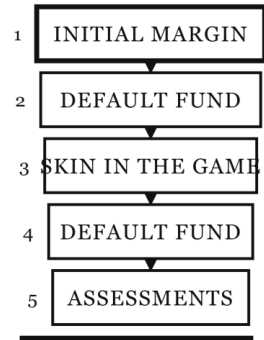
regulation is applied, and compliance considerations in cross-border transactions. It provides essential knowledge for compliance, legal, and risk management professionals, and for anyone whose business crosses borders.

Chapter 5. “Risk Management in Clearing”: Risk

management is the heart of the clearing system. This chapter identifies the diverse risks that CCPs and clearing members must manage — counterparty credit risk, market risk, liquidity risk, operational risk, legal risk, and systemic risk — and examines in professional depth how they are assessed and controlled. In particular, it details the default waterfall, the core mechanism underpinning a CCP’s financial soundness (the order in which losses are

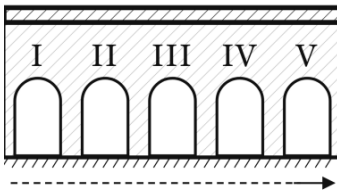
absorbed), and the conditions for its activation. It also explains the methodologies for calculating margin — initial margin and variation margin — which sit at the core of credit risk management (SPAN® and VaR-based models, among others), together with issues such as model accuracy and procyclicality (the correlation with market volatility), as well as the importance of stress testing and liquidity risk management (funding and eligible collateral). For risk management specialists and derivatives practitioners alike, this chapter offers deep insight from both theory and practice.

RISK MANAGEMENT



Chapter 6, “Onboarding and Documentation”: What procedures must a client complete, and what contracts must it sign, before it can actually use clearing services? This chapter focuses on client onboarding and the accompanying documentation — topics that may seem unglamorous but are critically important in practice. It

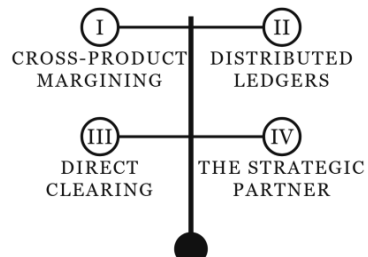
ONBOARDING AND DOCUMENTATION



concretely explains the clearing broker’s client eligibility reviews (KYC/AML), account opening procedures, preparations for system connectivity, and the principal agreements concluded with clients — clearing agreements, collateral agreements, and customer agreements — including their typical provisions and the points to watch in negotiation. A distinctive chapter offering practical knowledge for client-facing, operations, and legal and compliance staff, and indeed for clients using clearing services themselves.

Chapter 7, “The Future of Client Clearing”: Building on all that precedes it, this final chapter sketches how clearing — and client clearing in particular — may evolve from here. How will clearing brokers transform from mere intermediaries into strategic partners? Will integration with prime brokerage deepen? Will newer models such as sponsored access and direct clearing take hold? And how will technologies such as DLT and AI redraw the clearing landscape? The chapter considers these questions in light of the latest market developments and the lessons of past market events. It also analyzes what these

THE FUTURE OF CLIENT CLEARING



changes mean for the market as a whole, explores paths toward safer and more efficient market infrastructure, and offers the author's own recommendations on the challenges facing Japan's financial system. It is intended to be thought-provoking for anyone interested in where the market is headed — senior management and strategy professionals in particular.

This book covers clearing comprehensively, from fundamentals to advanced topics and on to the future. First-time readers will gain a systematic understanding by reading from Chapter 1 onward, while readers with specific interests may start with the chapter most relevant to them. The chapters are interrelated but written so that each can be read on its own, and the columns placed throughout the text offer deeper dives into particular topics.

Who This Book Is For

This book is intended for a broad readership, including:

- Practitioners at financial institutions engaged in derivatives trading, risk management, operations, compliance, and IT systems
- Those involved with market infrastructure at regulatory authorities, central banks, exchanges, and CCPs
- Students and researchers in finance
- Consultants, lawyers, accountants, and other professionals with an interest in clearing

Through this book, readers will be able to organize fragmentary knowledge of derivatives clearing into a coherent whole and grasp the full picture. They will come to understand the latest developments and the key issues in this fast-moving field, and find hints for solving the problems they face in daily practice and research. Beyond that, the book aims to provide — through the lens of clearing — deeper insight into risk management and regulation in financial markets, and into the future that technology will bring.

WHO THIS BOOK IS FOR



Welcome to the World of Clearing

Derivatives clearing may at first appear complex and forbiddingly specialized. Yet to understand how it works is to touch both the dynamism that drives modern financial markets and the collective ingenuity that keeps them stable — an intellectually exhilarating pursuit. If this book can serve as a compass to the evolving world of derivatives clearing, deepening readers' understanding and aiding their work and research, nothing would give me, as its author, greater joy. Now then — let us set out together to explore this profound world of clearing, so rich in possibilities for the future.

Chapter 1: The Fundamentals of Clearing

Section 1: Into the World of Clearing — The “Invisible Hand” Behind a Colossal Market

Modern financial markets form a complex and dynamic world in which vast sums of money move across borders in an instant. Within it, the market for derivatives (financial instruments whose value derives from underlying assets) underpins the very foundations of economic activity, serving as an indispensable tool for hedging risk and implementing investment strategies. Its scale is nothing short of astronomical: over-the-counter (OTC) derivatives alone reach approximately **USD 845 trillion** in notional amount, the contractual measure of outstanding exposure. That is roughly seven times the world’s annual GDP (2025 world GDP: approximately USD 118 trillion) — a figure that speaks to how central a role derivatives play in modern finance.

Yet for this vast and complex market to function smoothly day after day, an elaborate apparatus must operate behind the scenes — a “market infrastructure” that ensures the “promises” embedded in every trade are honored and that safeguards the stability of the market as a whole. **Clearing**, the subject of this book, sits at the very core of that market infrastructure.

What exactly is clearing? And why is it indispensable to a derivatives market of such enormous scale?

This chapter, “The Fundamentals of Clearing,” is the gateway to answering these foundational questions. We begin by establishing that every financial transaction follows a lifecycle made up of a sequence of stages — trading, clearing, and settlement — and by clarifying where clearing fits within that flow. Next, we explain the basic concepts of clearing, drawing on familiar analogies, and unpack the role of the margin system (IM/VM) that lies at its core. We then examine the key benefits clearing

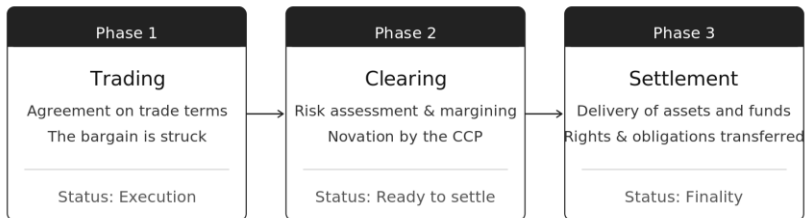
delivers — risk reduction, greater efficiency, and improved transparency — and introduce the principal players that sustain this ecosystem: central counterparties (CCPs), clearing brokers, and clients. Finally, we survey how clearing has evolved historically and, in particular, how it arrived at its present form through the crucible of the 2008 global financial crisis.

By the end of this chapter, readers will have grasped the basic structure and significance of what may at first appear to be a bewilderingly complex world, and will have built a solid foundation for the more detailed discussions developed in the chapters that follow — cleared products, risk management, laws and regulations, operational processes, and the outlook for the future. Let us now begin our exploration of clearing: the “invisible hand” that supports the world’s vast financial markets from behind the scenes.

1. Introduction: The Lifecycle of Financial Transactions and the Importance of Clearing

1.1 The Three Stages Underpinning Financial Transactions: Trading, Clearing, and Settlement

In today’s financial markets, an immense volume of transactions takes place every day across a wide range of instruments — equities, bonds, and derivatives, the central subject of this book. For these transactions to proceed smoothly and safely, a sequence of processes — the “lifecycle” of a financial transaction — must function properly. This lifecycle can be divided broadly into the following three stages.



1. Trading:

- **Substance and role:** This is the stage at which a seller and a buyer agree on the terms of a particular financial instrument — price, quantity, settlement date, and so on. It can be thought of as the act of exchanging a “promise” concerning future actions.
- **Example:** Investor A promises Investor B to “sell 100 shares of a particular company’s stock at JPY 1,000 per share, three days from now.” In a derivatives transaction, the parties might enter into a contract such as “three months from now, pay a floating rate of interest in exchange for receiving a specified rate.”
- **Status:** At this stage, no actual assets (shares or funds) have yet changed hands. Only the “promise” has been established.

2. Clearing:

- **Substance and role:** This is the stage at which preparations are made to ensure that the “promise” embodied in the trade will be performed without fail, and at which the associated risks are managed. It involves confirming the trade details, assessing risk, and calculating and managing the required collateral (margin). In derivatives transactions in particular, clearing is a critically important process for managing the risk that one’s counterparty becomes unable to honor its obligations (counterparty risk).
- **Example:** For an equity trade, the security traded, the quantity, the price, the settlement date, and other details are verified for accuracy. For a derivatives transaction, in addition to these checks, the potential future losses that could arise from the trade are assessed, and the amount of margin required to cover that risk is calculated, collected from the parties, and managed.
- **Status:** At this stage, concrete preparations for the performance of the trade and the associated risk management are carried out. Where a central counterparty (CCP) is interposed, the CCP becomes the counterparty to the trade and assumes the role of guaranteeing the performance of settlement (discussed in detail below).

3. Settlement:

- **Substance and role:** Based on the details finalized in the clearing process, this is the final stage at which assets — shares, bonds, funds, and so on — are actually delivered. It is the stage at which the promise is “executed.”
- **Example:** In an equity trade, the shares are transferred from seller to buyer, and the purchase price is paid by the buyer to the seller. In a derivatives transaction, either the net difference calculated under the contract is paid (cash settlement) or the underlying asset is delivered (physical delivery).
- **Status:** At this stage, the transaction is fully completed, and the transfer of legal rights and obligations becomes final.

1.2 Why Does Clearing Matter? The Linchpin of Financial Market Stability and Efficiency

Of these three stages, it is clearing on which this book focuses. Why does clearing play such a pivotal role in modern financial markets, rather than being mere “intermediate processing” between trading and settlement? Because clearing provides three essential functions that underpin the very foundations of financial markets.

1. Risk Reduction	Ensuring safety
The CCP becomes the central counterparty to all participants, severing bilateral credit risk	
• Elimination of counterparty risk • Collateralization through the margin system • Loss-mutualization scheme on default	
2. Trading Efficiency	Smoother trading
Netting sharply reduces the actual movement of cash and securities needed for settlement	
• Multilateral netting • Greater funding and liquidity efficiency market-wide • Lower operational burden and settlement costs	
3. Greater Transparency	Sounder markets
Trade information is centralized at CCPs and trade repositories (TRs), making risk visible	
• Visibility and centralized monitoring of risk • Restraint on excessive risk-taking • Macroprudential assessment by regulators	

1. Risk reduction (ensuring safety): Financial transactions — particularly long-dated, complex derivatives transactions — always carry counterparty risk: the risk that the other party to the trade becomes unable to pay. Clearing, and in particular clearing through a CCP, reduces this risk dramatically. By standing at the center of the trade and becoming the counterparty to both the buyer and the seller, the CCP shields market participants from the credit risk of individual counterparties. Moreover, through mechanisms such as the margin system (discussed below), it rigorously manages price risk and default risk, providing for losses should they ever

materialize.

2. Greater efficiency (smoother trading): Clearing also contributes substantially to the efficiency of the trading process. By aggregating large numbers of trades and offsetting claims and obligations against one another (netting), a CCP can dramatically reduce the amounts of funds and securities that actually need to be settled. This lowers settlement costs and operational burdens and improves the funding efficiency of the market as a whole. Standardization of the trading process likewise speeds up transactions and reduces errors.

3. Improved transparency (a sounder market): When trade information is aggregated at, and reported to, CCPs and trade repositories (TRs), market participants and regulators gain a far more accurate picture of risk conditions across the market. Making the location of risk visible is essential to preserving the soundness of the market and restraining excessive risk-taking.

1.3 The Focus of This Book: Into the Depths of Derivatives Clearing

Many excellent specialist works already cover the mechanics of derivative products, trading strategies, and pricing theory. Yet there have been few attempts — particularly in the Japanese-language literature — to focus specifically on clearing, the “safety net” of derivatives trading and the core of market infrastructure, and to explain its mechanics, risk management, legal and regulatory framework, and latest developments in a systematic and practical manner.

This book aims to shine a light on derivatives clearing — a specialized yet critically important field — and to unravel its complexity and depth. It provides comprehensive coverage, from basic concepts to the most advanced debates, of a field whose importance has grown dramatically, and which has evolved rapidly, since the 2008 global financial crisis.

Clearing is not merely a piece of back-office processing. It is a dynamic and

intellectually demanding domain that calls for sophisticated risk management techniques, compliance with complex laws and regulations, and the deployment of cutting-edge technology.

Beginning with the next section, we explore the basic concepts of clearing through familiar analogies, to answer the question: how exactly does clearing work?



2 What Is Clearing? A Safety Net for the “Promises” of Financial Transactions

2.1 A Familiar Analogy: A Promise to Exchange Fixed and Floating Interest Rates (an Interest Rate Swap)

As noted in Section 1.1, clearing plays the vital role within the lifecycle of a financial transaction of managing risk and securing the safety of the trade. The term “clearing,” however, may sound somewhat technical. To grasp its essence, let us begin with the simplest possible illustration, using an interest rate swap — one of the most representative derivatives. Suppose that you — let us call you “B” for convenience — have borrowed money at a floating rate of interest. If rates rise in the future, you run the risk that your interest payments will increase. Your friend A, meanwhile, has surplus funds and would like to invest them at a floating rate, but worries about the risk that interest income will shrink if rates fall.

A and B therefore decide to enter into the following **interest rate swap agreement**.

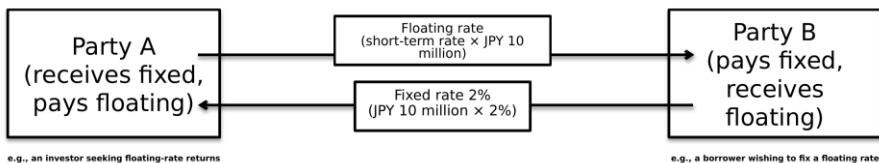
■ **Notional amount:** JPY 10 million

■ **Term:** one year (to keep the illustration simple)

■ **B:** pays a fixed rate of 2% and, in exchange, receives the floating rate

■ **A:** pays the floating rate and, in exchange, receives a fixed rate of 2%

Basic Structure of an Interest Rate Swap



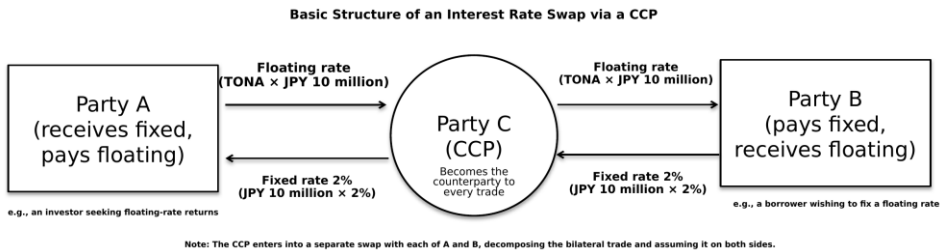
Note: The notional amount (JPY 10 million) does not actually change hands.

The notional amount of JPY 10 million never changes hands itself, but depending on the future level of interest rates, cash flows will arise that favor one party or the other.

This uncertainty about future interest rates and cash flows is the source of counterparty risk in derivatives transactions.

Here, A holds the “right to receive the fixed 2%” and B bears the “obligation to pay the fixed 2%,” and there is a risk that one of the two becomes unable to keep the promise partway through (that is, defaults). If, for example, market rates fall sharply and the trade moves against B, B might stop making payments. This is where **C** enters the picture — a mutual friend of the two who, let us assume, enjoys exceptionally strong credit. C steps in between them and makes the following proposal.

“A, B — rather than making promises directly to each other, **let me become the counterparty to each of you**. I will receive the fixed 2% from B and pay the fixed 2% to A. Likewise for the floating rate: I will stand in the middle and intermediate the



payments between you.”

The role that C plays here is performed in real financial markets by the **central counterparty (CCP)**. By interposing itself between the parties to a trade and becoming the counterparty to each, the CCP assumes counterparty risk and guarantees the performance of the trade — it is, quite literally, the “trusted intermediary” of the financial markets.

2.2. A Concrete Scenario: Enter Margin (IM and VM)

How, then, does the CCP (our friend C) actually manage risk and guarantee the performance of trades? This is where the crucial mechanism of **margin** comes into play. Margin functions as a “performance bond” and as a “daily settling-up payment” that secures the safety of the trade.

Initial Margin (IM)

Purpose: Before a trade begins, IM is posted as a “performance bond” against the possibility of a failure to perform (a default). The CCP calculates IM by reference to whether, should the counterparty default, this margin would be sufficient to absorb the losses expected over a given period.

Process: The CCP (C) **requires both parties — B, who pays the fixed rate, and A, who pays the floating rate — to post a set amount of initial margin.**

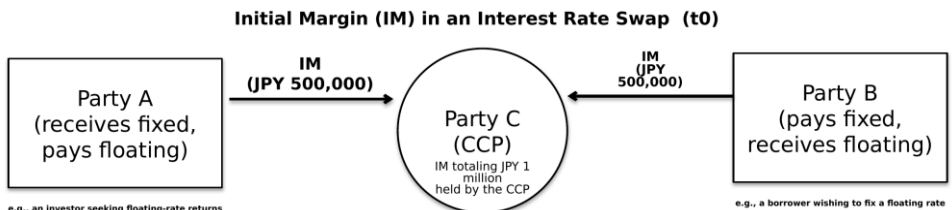
Suppose, for example, that for an interest rate swap with a notional of JPY 10 million, an assessment based on historical interest rate data concludes that a loss of up to roughly JPY 500,000 could arise over the several days required to complete close-out. In that case, the CCP requires B and A each to post **JPY 500,000** as IM. This is the initial margin (IM).

Why both parties?

In an interest rate swap, depending on where interest rates go, sometimes the fixed-rate payer (B) loses money, and sometimes the floating-rate payer (A) does.

- If interest rates rise sharply, A — the party “receiving” the fixed 2% — tends to be at a disadvantage
- If interest rates fall sharply, B — the party “paying” the fixed 2% — tends to be at a disadvantage

Neither party is permanently the “lender” or the “creditor”; rather, **depending on how prices move, either side may suffer a loss.** For this reason, it is standard practice in



At this point, the mark-to-market gain or loss is assumed to be zero, and only IM has been posted.

derivatives transactions for both parties to post IM. The CCP thereby secures a performance bond sufficient to cover losses from price movements over a given period, whichever side defaults.

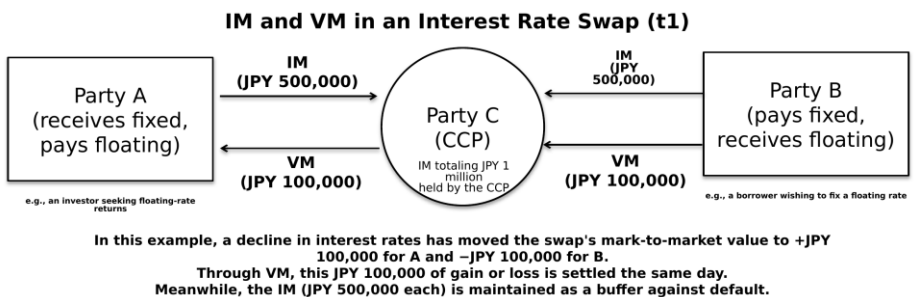
Variation Margin (VM)

Purpose: VM is the cash that settles **the mark-to-market value of the swap (its unrealized gain or loss)** arising from day-to-day movements in market interest rates after the trade begins. It prevents losses from accumulating to excessive levels.

Process: Once the trade is under way, market interest rates move every day. As a result, the mark-to-market value of the interest rate swap between A and B changes from moment to moment.

Suppose, for example, that some time after inception, interest rates have fallen, creating an environment favorable to the party receiving the fixed 2% (A). If the swap were terminated at market value at this point, A would be assessed to have **a gain of JPY 100,000** and B **a loss of JPY 100,000**.

VM is what settles, that same day, this “JPY 100,000 of gain or loss that would crystallize upon termination today.” The CCP (C) collects **JPY 100,000** from the losing side (here, B) and pays it to the gaining side (A). (If rates move again on subsequent days, VM may flow in the opposite direction, based on the market value prevailing at that time.)



2.3 The Cash That Actually Moves, and How to Think About Gains and Losses

The crucial point here is that the funds actually moving from day to day are not the JPY 10 million notional itself, but only the IM (JPY 500,000 from B plus JPY 500,000 from A, JPY 1 million in total) and the VM corresponding to the mark-to-market gain or loss at the time.

Let us revisit the example illustrated in the figures “IM in an Interest Rate Swap” and “IM and VM in an Interest Rate Swap.”

First, at trade inception (to), A and B have each deposited JPY 500,000 with the CCP as IM.

At this point, market interest rates are assumed to be the same as the rate at which the trade was struck, so the mark-to-market value of the swap (its unrealized gain or loss) is zero. The only money actually in motion is therefore the combined IM of JPY 1 million.

Now suppose that interest rates subsequently fall, moving the market in a direction favorable to A, who receives the fixed 2%.

Suppose that, if the swap were terminated at market value at this point,

- A would record a gain of +JPY 100,000
- B would record a loss of -JPY 100,000

Let us consider the case assessed in this way.

The role of VM is to settle, that same day, this “JPY 100,000 of unrealized gain or loss that would crystallize upon termination today.”

The CCP collects JPY 100,000 as VM from B, the party carrying the unrealized loss, and pays JPY 100,000 as VM to A, the party holding the unrealized gain.

At this point, each participant’s position can be summarized as follows.

• B:

B has posted JPY 500,000 as IM and paid JPY 100,000 as VM. This amounts to a total cash outflow of JPY 600,000, but in exchange B keeps alive the swap contract under

which cash flows will continue to be exchanged in line with future interest rate movements.

•A:

A has posted JPY 500,000 as IM and received JPY 100,000 as VM.

On a net basis this is a cash outflow of JPY 400,000, but the mark-to-market value of the swap has moved JPY 100,000 in A's favor. If the trade were unwound at this point, the IM would in principle be returned, so A would have realized the JPY 100,000 of profit received as VM.

In this way, an interest rate swap transaction does not move the large JPY 10 million notional itself; rather, day-to-day risk and profit and loss are settled through comparatively compact exchanges of cash in the form of IM and VM.

IM is maintained throughout the life of the trade as a “buffer against a possible default,” while VM functions as a “mechanism for settling each day's mark-to-market gain or loss immediately” — keeping these two roles distinct is essential to understanding how clearing works.

Further Changes in Value and the CCP's Guarantee Function

Now consider what happens if, on the following day, interest rates fall further and the mark-to-market value of the swap moves still more in A's favor.

As of the previous day (t_1), the unrealized gain or loss stood at +JPY 100,000 for A and -JPY 100,000 for B.

Suppose rates then fall further, and the position is now assessed, relative to inception (t_0), at an unrealized gain of +JPY 1,000,000 for A and an unrealized loss of -JPY 1,000,000 for B.

In this case, the newly arising unrealized gain or loss — the “increment since the previous day” — is:

$$+JPY 1,000,000 - (+JPY 100,000) = +JPY 900,000 \text{ (for A)}$$

$$-JPY 1,000,000 - (-JPY 100,000) = -JPY 900,000 \text{ (for B)}$$

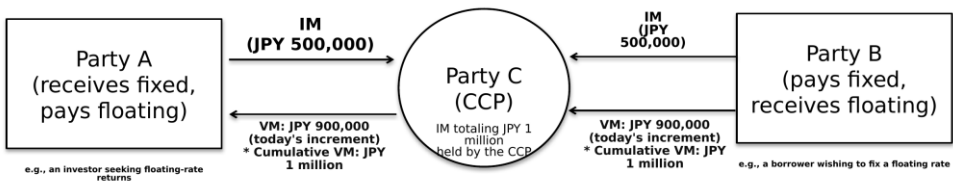
For this increment of JPY 900,000, the CCP collects JPY 900,000 from B and pays JPY 900,000 to A as variation margin (VM).

Aggregating the movements over these two days, VM has flowed as follows.

- Day 1 (t₁): VM of JPY 100,000 (B → A)
 - Day 2 (t₂): VM of JPY 900,000 (B → A)
- In total, JPY 1,000,000 has been transferred from B to A as VM

Meanwhile, the IM (JPY 500,000 from A plus JPY 500,000 from B, JPY 1 million in total) remains on deposit with the CCP at all times. The division of labor is clean: IM serves as the “safety valve against a default,” while VM is the “mechanism that settles the mark-to-market gain or loss day by day.”

IM and VM in an Interest Rate Swap (t₂)



On a cumulative basis from t₀, the swap's mark-to-market value is +JPY 1 million for A and -JPY 1 million for B. Over the two days, VM totaling JPY 1 million (JPY 100,000 on day 1 + JPY 900,000 on day 2) has flowed from B → CCP → A.

The CCP functions as an “intermediary,” receiving VM from B and paying VM to A. Even if A were to become unable to pay, the CCP bears the responsibility of fulfilling the payment to B, using the IM it holds, its own capital, and the default fund (the default waterfall).

Now imagine that you are A, trading directly (bilaterally) with B. You might well worry whether B, carrying an unrealized loss, can truly keep paying VM — or whether B might one day simply stop paying altogether.

Under CCP clearing, however, your counterparty is not B but the CCP. The CCP calls and manages VM and IM vis-à-vis B while, at the same time, providing you with a performance guarantee: the VM determined to be due to you will be paid, without fail. Even if B were to become unable to pay, the CCP bears the responsibility of making the

payments due to you, drawing on the IM it holds from B and, where necessary, on its own capital, the default fund, and other resources (the so-called default waterfall, taken up in detail in Chapter 5).

This is the core function of CCP clearing: it lets participants focus on trading without undue concern about the credit risk of their counterparties (counterparty risk).

2.4. The Role of the CCP: The Foundation of Market Confidence and Stability

As we have seen through the concrete interest rate swap example and the mechanics of margin, the CCP plays an extraordinarily important role in financial markets.

●**Assumption of counterparty risk**: By interposing itself between the parties to a trade and becoming the counterparty to each, the CCP concentrates and assumes the counterparty risk that individual participants would otherwise bear.

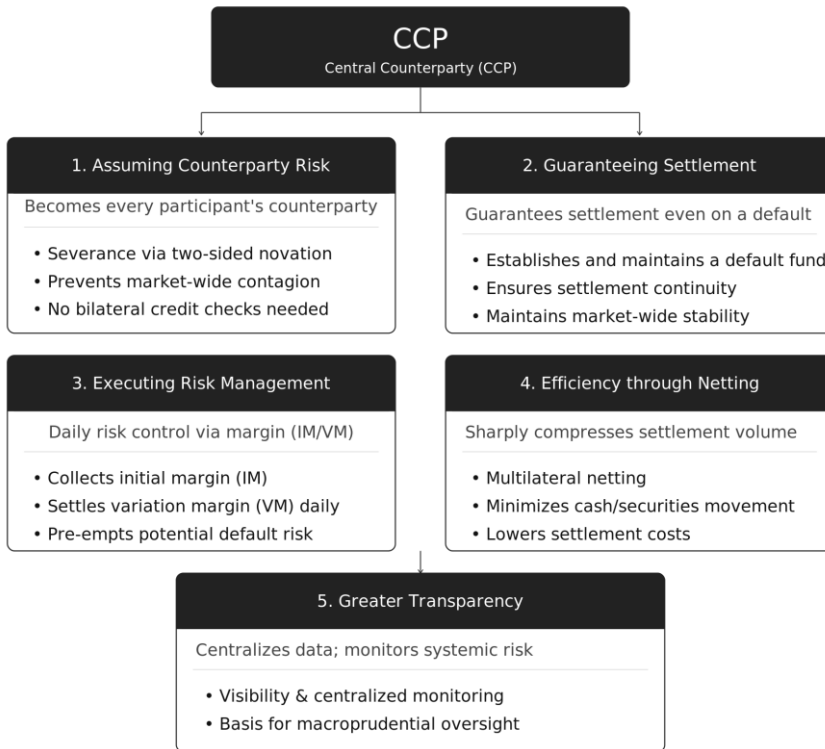
●**Guarantee of settlement performance**: Even if a trading participant defaults, the CCP guarantees the performance of settlement, preventing disruption from spreading across the market.

●**Execution of risk management**: The CCP operates the margin system (IM/VM) rigorously, managing day-to-day price risk and default risk.

●**Efficiency through netting**: By aggregating large numbers of trades and offsetting them against one another (netting), the CCP reduces settlement volumes and administrative costs, enhancing the efficiency of the market as a whole.

●**Improved transparency**: By aggregating and managing trade information, the CCP makes it easier to grasp the risk profile of the market as a whole, thereby enhancing transparency.

Through these functions, the CCP provides the indispensable foundation on which financial markets can operate more **safely**, more **transparently**, and more **efficiently**. In Japan, the Japan Securities Clearing Corporation (JSCC) performs this vital role; in Europe and the United States, it is performed by LCH, CME Clearing, and others, each



in its own market.

In the next section, we look in greater detail at the margin system, the core of the clearing mechanism.

3 The Mechanics of Clearing: Risk Management Through Margin

In Sections 2.2 and 2.3, we used a concrete interest rate swap example to see how initial

margin (IM) and variation margin (VM) are posted and how they move from day to day. In this section, we step back a little and consider **the role these margins play as the core of risk management in clearing.**

The risk a CCP assumes is the loss that could arise while closing out or hedging a position in the market should one of the trading participants default. The margin system can be understood as a mechanism that decomposes this loss into two components along the time axis and manages each of them.

- Providing for the “potential losses” that could arise over a given future period is the role of **initial margin (IM)**
- Settling, as it arises, the “gains and losses already incurred” through actual market price movements after the trade begins is the role of **variation margin (VM)**

Such is the division of labor. IM is a buffer against losses that “have not yet occurred but could,” while VM is the settlement mechanism that clears gains and losses that “have already occurred.”

Because losses can arise on either side of a derivatives transaction, **both parties ordinarily post IM**, and once the trade is under way, **each day’s gains and losses are exchanged as VM** in line with price movements. The result is a two-layered approach to risk management: a buffer (IM) sufficient to cover losses up to a certain level is set aside in advance in case a default occurs, while daily gains and losses are settled promptly through VM, preventing losses from snowballing.

In what follows, we **set out the respective roles of IM and VM from a more general, conceptual standpoint** (the specific calculation models, parameter settings, scenario design, and other details are treated at length in Chapters 2 and 5).

3.1 Initial Margin (IM):

A Buffer Against “Future Losses”

Initial margin (IM) is **a buffer posted in advance to cover the losses that could**

result from “the worst price moves that might occur over a given future period”.

When a trade is opened, the CCP requires each participant to post IM commensurate with the riskiness of its positions.

The logic behind IM can be broken down a little further, as follows.

- The CCP envisages “the losses that could arise — should a participant suddenly default — over the time needed to hedge or close out that participant’s positions in the market.”
- This “period required for hedging and close-out” is set as the margin period of risk (MPOR) — for example, one to two days for listed futures and five to ten days for OTC derivatives.
- Using historical market data and stress scenarios, the CCP estimates statistically “how large a loss could arise over the MPOR within the range it ordinarily assumes,” and sets the IM level so as to cover that amount.

With this picture in mind, **IM** can be understood as **collateral intended to cover, up to a prescribed confidence level, the losses that may arise between the last VM payment (that is, the last mark-to-market) and the moment the CCP finishes closing out or hedging the position.**

In practice, IM is calculated using models (such as SPAN® or VaR-based models) that take into account the notional amount of the position, its remaining maturity, volatility, and correlations (the diversification effect of the portfolio as a whole). These details are taken up in Chapters 2 and 5; for now, it is enough to grasp that IM is a **shield against the “future” losses that would follow a default** and that — reflecting the two-way loss potential peculiar to derivatives — **both sides of a trade are ordinarily required to post IM.**

3.2 Variation Margin (VM):

A Mechanism for Resetting “Daily Gains and Losses”

Variation margin (VM) is the mechanism by which **the gains and losses that have already arisen from actual market price movements after a trade begins are settled daily, or even intraday.**

As the interest rate swap example showed, once a trade is open, the swap’s mark-to-market (MTM) value fluctuates daily with movements in interest rates. When the swap’s value rises, one party records an unrealized gain and the other an unrealized loss. Left in place, these valuation gains and losses would steadily pile up on the losing side — until, one day, a default suddenly occurs with large unsettled losses outstanding. To address this, every business day — and in some cases several times a day, intraday included — the CCP typically:

1. Calculates the latest market value of each position;
2. Computes how much each valuation has moved since the previous mark-to-market (the unrealized gain or loss);
3. Transfers that difference as VM — from the losing side to the CCP, and from the CCP to the gaining side.

This cycle ensures that **past gains and losses are “settled” in cash as they arise, leaving only the losses that could emerge over “the short interval since the last mark-to-market”** — a state the CCP maintains at all times.

This point is critical to understanding the relationship between IM and VM.

- Thanks to VM, past losses are paid as they occur, so that “a mountain of unsettled losses” is unlikely to build up.
- Consequently, what the CCP faces upon a default is **the loss arising over “the period since the last VM payment”** — and IM is designed to cover precisely that portion.

Put differently,

VM plays the role of “clearing away, as they arise, the gains and losses that have already occurred,”

while IM plays the role of “providing against the losses that may yet occur.”

The combination of these two functions is what underpins the stability of CCP clearing.

In practical terms, VM can fairly be called “the centerpiece of the daily cash flows in exchange and CCP clearing”; because it **bears directly on the liquidity management of clearing participants and clients (the sourcing of cash and collateral)**, it is examined in detail in Chapters 3 and 5.

4 Why Does Clearing Matter? Contributions to Market Stability and Efficiency

Section 1.3 established that the margin regime is the core of risk management in clearing. Through this rigorous risk-management mechanism, clearing does more than enhance the safety of individual trades: it makes several essential contributions to the stability and efficient operation of the financial markets as a whole. This section revisits the principal benefits clearing delivers and explores why they matter.

Importance of Clearing



- Risk Mitigation**
Mitigates counterparty risk and ensures market stability.
- Operational Efficiency**
Standardizes transaction processes and reduces costs.
- Enhanced Transparency**
Centralizes transaction information and increases market confidence.

4.1 The Risk-Mitigation Role (Ensuring Safety)

The most fundamental and important role of clearing is to reduce dramatically the risks inherent in financial transactions — above all, **counterparty risk**.

● **Trade guarantee:** By standing at the “center” of each trade — becoming the seller to

every buyer and the buyer to every seller — the CCP guarantees performance even when an individual counterparty defaults. Market participants can therefore trade with confidence, free of undue concern about each counterparty’s creditworthiness. History has shown, in crises such as the 2008 global financial crisis, just how vital this function is.

●**Protection through the margin regime:** As we saw in Section 1.3, initial margin (IM) serves as a “performance bond” against losses in the event of a default, while variation margin (VM) serves as a “daily settlement payment” that clears the gains and losses arising from day-to-day price movements. This two-layered margin regime provides for potential losses while preventing them from accumulating to excess.

●**Default fund (DF):** In addition, the CCP maintains multi-layered financial resources — the default waterfall — comprising funds contributed by clearing participants (the default fund) and the CCP’s own capital (its skin in the game, or SIG). Even where losses exceed a participant’s margin, these resources stand ready to keep the impact on the broader market to a minimum. (Default management is covered in detail in Chapter 5.)

4.2 The Efficiency Role (Smoother Trading)

Clearing also contributes substantially to the efficiency of the trading process, reducing costs and operational burdens for market participants.

■ **Multilateral netting:** Because the CCP concentrates trades from many market participants, the multiple claims and obligations each participant holds can be offset — netted — on a multilateral basis. This sharply reduces the amounts of cash and securities that actually have to be settled, cutting settlement costs and operational workloads. And because risk exposures themselves are netted, the amount of margin required falls as well. (Details are provided in Chapter 3.)

- **Standardization of the trading process:** In clearing, the processes from trade matching through margin calculation and exchange to settlement are standardized. The result is faster trading, fewer errors, and more efficient system connectivity.
- **Centralized margin management:** However many counterparties a participant faces, margin is posted centrally to the CCP (or to the CCP via a clearing broker), which greatly simplifies its management.

4.3 Greater Transparency (A Sounder Market)

Clearing also serves to enhance transparency across the market and to promote sounder market operation.

- **Centralized management and reporting of trade data:** Most OTC derivatives trades must be reported to CCPs and to trade repositories (TRs). This allows regulators to grasp market-wide trading activity and the location of risk far more accurately, and to exercise appropriate supervision.
- **Clear risk-management standards:** CCPs publish their margin calculation models and risk-management rules, enabling market participants to assess a CCP's risk-management standards objectively. This raises confidence in the market as a whole.

Why Clearing Underpins the Market

Through three vital pillars — risk mitigation, trading efficiency, and greater transparency — clearing provides the indispensable foundation on which modern financial markets function safely and smoothly. In derivatives markets especially, where transactions are complex and risks run high, the CCP-centered clearing system plays a decisive role in preserving market stability and confidence.

5 The Key Players in Clearing

The discussion so far has shown that clearing — and the CCP in particular — plays a central role in market stability and efficiency. Yet the clearing framework does not rest on the CCP alone. It involves several players, each with a distinct role, whose interconnections form what may be called the “clearing ecosystem.”

This section briefly introduces the principal players that make up this ecosystem. (Their specific roles and interrelationships are explored in greater depth in Chapter 3.)

5.1 Central Counterparty (CCP)

- **Role:** The core of the clearing system. As a “trusted intermediary,” the CCP interposes itself between the parties to a trade, assumes counterparty risk, and guarantees settlement. It performs a wide range of functions that support market-wide stability, including margin calculation and management, risk monitoring, and default management.
- **Examples:** Japan Securities Clearing Corporation (JSCC), LCH, and CME Clearing.

5.2 Clearing Participants (Clearing Members / Clearing Brokers / FCMs)

- **Role:** Financial institutions — chiefly banks and securities firms — that connect directly to the CCP and are qualified to use its clearing services. Beyond clearing their own trades, they also act as “clearing brokers,” intermediating clearing for other market participants (clients) that cannot access the CCP directly. They also manage client risk and post margin with the CCP on clients’ behalf, among other functions.
- **Examples:** Major banks and major securities firms.

5.3 Clients (End Users)

- **Role:** The market participants who actually trade derivatives, whether to hedge

risks or to invest. Often unable to access the CCP directly, they use clearing services indirectly through a clearing participant (clearing broker).

○**Examples:** Asset managers, hedge funds, pension funds, insurance companies, and non-financial corporations.

With each of these players performing its role, the clearing system functions smoothly, underpinning the stability and efficiency of the financial markets as a whole.

6 The Evolution of Clearing: Historical Background and the Link to the Present

Clearing is by no means a recent invention. The need to manage risk in financial transactions has long been recognized, and history offers several early prototypes. At the Dojima Rice Exchange in eighteenth-century Japan, for example, the book-entry trading known as “chōai” featured cash settlement without physical delivery — devices that clearly anticipate modern clearing (see the column for details).

The modern clearing regime, however — above all the CCP-centered framework for the over-the-counter (OTC) derivatives market — developed and strengthened rapidly only in the wake of the unprecedented financial crisis of 2008. That crisis laid bare the fragility of an OTC market in which regulatory oversight fell short and counterparty risk propagated through opaque chains.

Drawing on those grave lessons, the international community embarked, under the agreements reached at the G20 summits, on a root-and-branch reform of the OTC derivatives market. At its core were mandatory central clearing through CCPs, the promotion of electronic trading platforms, trade reporting obligations, and the introduction of margin requirements for non-cleared derivatives (see Chapter 4 for details).

In this way, the modern clearing regime — while resting on the accumulated wisdom

of historical risk management — has evolved dramatically, shaped by the lessons of the financial crisis and the tide of international regulatory reform, into critical infrastructure for securing market stability and transparency.

Section 1 in Review

- Financial transactions follow a lifecycle of trading, clearing, and settlement; clearing sits at its core, providing risk management and guaranteeing performance.
- Clearing is a mechanism in which the CCP becomes the counterparty to every trade and manages counterparty risk through the margin (IM/VM) regime.
- Clearing delivers important benefits: risk mitigation, trading efficiency (netting and the like), and greater market transparency.
- The clearing ecosystem comprises CCPs, clearing participants (the sell side and clearing brokers), and clients.
- The modern clearing regime has evolved dramatically through the lessons of the 2008 global financial crisis and international regulatory reform.

Section 2 Historical Background and Evolution: The Relentless Pursuit of Risk Management

1 The Origins of Clearing: Early Experiments in Risk Management

As noted in Section 1.6, clearing developed over many years out of the need to manage the risks of financial transactions and to bolster confidence in markets. It did not appear fully formed overnight; rather, it is the cumulative product of ingenuity applied in markets around the world to the challenges of each era. Here we trace the early seeds of the clearing function — the precursors of today’s CCPs — through several historical markets. (For the detailed history and mechanics of each market, see the separate columns.)

1.1 The Dojima Rice Exchange (Eighteenth-Century Japan): The Seeds of Futures Trading and Clearing

At the Dojima Rice Exchange in eighteenth-century Japan, regarded as one of the world’s earliest organized futures markets, sophisticated arrangements emerged for managing the risk of rice price fluctuations. In particular, the futures trading known as “chōai-mai” (book-entry rice trading) widely employed cash settlement of book differences alone, without any physical delivery of rice. Moreover, the clearing function performed by the “keshiai-ba” and the settlement-guaranteeing intermediaries known as “yarikuri ryōgae” can be seen as prototypes of the modern CCP and the clearing membership system — strikingly advanced exercises in risk management.

1.2 The London Stock Exchange (Nineteenth-Century Britain): The Battle Against Credit Risk

At the nineteenth-century London Stock Exchange (LSE) — by then the world’s financial center under the Industrial Revolution and the British Empire — surging trading volumes made counterparty risk, the danger that a counterparty would fail to

pay on settlement day, a serious problem. In response, the LSE worked to streamline its settlement processes (a forerunner of netting) and developed arrangements requiring participants to post collateral (a precursor of modern margin). These measures embodied the basic philosophy of risk-based management: requiring greater safeguards for riskier trades and riskier participants.

1.3 The Chicago Mercantile Exchange (Twentieth-Century United States): The Establishment of Modern Clearing

The Chicago Mercantile Exchange (CME) in the United States, which began with agricultural commodities, developed a modern clearing system in the twentieth century as it extended its reach into financial derivatives. It established a clearing house early on and cemented its role as a CCP, becoming the counterparty to each trade and guaranteeing settlement. It also introduced and refined a precise margin regime built on initial margin (IM) and variation margin (VM); in particular, the SPAN® system (developed in 1988), which assesses risk across the portfolio as a whole, set the standard for risk management in derivatives markets worldwide. Its robustness proven even through the 2008 global financial crisis, CME's clearing system stands as one of the model cases for the modern CCP.

2 The Evolution of Clearing in Japan: The Founding of JSCC and the Road to Consolidation

In Japan, too, the clearing function has evolved alongside the development of the markets. The Tokyo Stock Exchange, the Osaka Securities Exchange, and other exchanges once each operated their own clearing functions, but as markets expanded and the demand for efficiency grew, momentum built to consolidate these functions in a specialized third-party institution.

The result was the establishment in 2002 of the **Japan Securities Clearing**

Corporation (JSCC). JSCC was created to secure the independence of the clearing business and to centralize risk management, thereby strengthening the stability and credibility of the market as a whole. After its founding, JSCC progressively expanded its role, integrating the clearing of equities, Japanese government bonds, and exchange-traded derivatives such as listed futures and options, and laying the foundations for its position as the central clearing house of Japan's capital markets. (For the background to JSCC's establishment and the subsequent integration of functions, see the column.)

3 The Turning Point: The 2008 Global Financial Crisis and OTC Derivatives Reform

The greatest turning point in the history of clearing came with the global financial crisis of 2008. This unprecedented crisis laid bare the fragility of the over-the-counter (OTC) derivatives market, which had been expanding rapidly while remaining largely out of regulators' sight.

A Turning Point: The 2008 Crisis and OTC Derivatives Reform

Bankruptcy of Lehman Brothers ◦ OTC-market fragility laid bare ◦ Counterparty risk surfaces all at once	CME Clearing & LCH SwapClear ◦ Used for listed futures & some OTC products ◦ Functioned through the crisis	G20 Pittsburgh Summit (2009) ◦ Central-clearing mandate for standard OTC ◦ Promote e-trading ◦ Trade reporting to TRs	G20 Cannes Summit (2011) ◦ Strict margin rules (UMR) for non-cleared ◦ Incentive to shift to CCP clearing
2008	2008	2009	2011

3.1 The Challenges Exposed by the 2008 Crisis

The bankruptcy of Lehman Brothers, a financial institution of enormous scale, tore through the web of counterparty exposures stretched across the OTC market and triggered acute credit anxiety among market participants. Opacity — no one knowing who held how much risk — amplified the panic, which escalated into systemic risk that paralyzed the financial system as a whole.

During the crisis, on the other hand, existing CCPs such as CME Clearing and LCH SwapClear functioned effectively: for derivatives that had become clearable (mainly listed products and certain interest rate swaps), they wound down the defaulted Lehman positions in relatively orderly fashion and helped contain the chain reaction

of risk.

3.2 The G20 Agreements and the Launch of Reform: From Pittsburgh to Cannes

This experience became the decisive catalyst for an international consensus that the efforts of individual financial institutions were not enough to manage the risks of the OTC market, and that centralized risk management through CCPs was indispensable. As a result, at the 2009 G20 Pittsburgh Summit, international agreement was reached on fundamental reform of the OTC derivatives market. Its core elements were ① **mandatory central clearing of standardized OTC derivatives through CCPs**, ② the promotion of trading on electronic platforms, and ③ trade reporting obligations (to TRs). The direction was thus clearly set: concentrate counterparty risk in CCPs and raise the transparency of the market.

Complementing these reforms, the 2011 G20 Cannes Summit added agreement on ④ **strict margin requirements for trades not centrally cleared (the obligation to exchange initial and variation margin — the so-called UMR)**. These rules strengthened risk management for non-cleared trades while also serving as an economic incentive to shift toward CCP clearing. (These regulations are explained in detail in Chapter 4.)

3.3 The Expansion and Evolution of the CCP's Role

The reforms flowing from the G20 agreements transformed clearing — and the role of the CCP — dramatically.

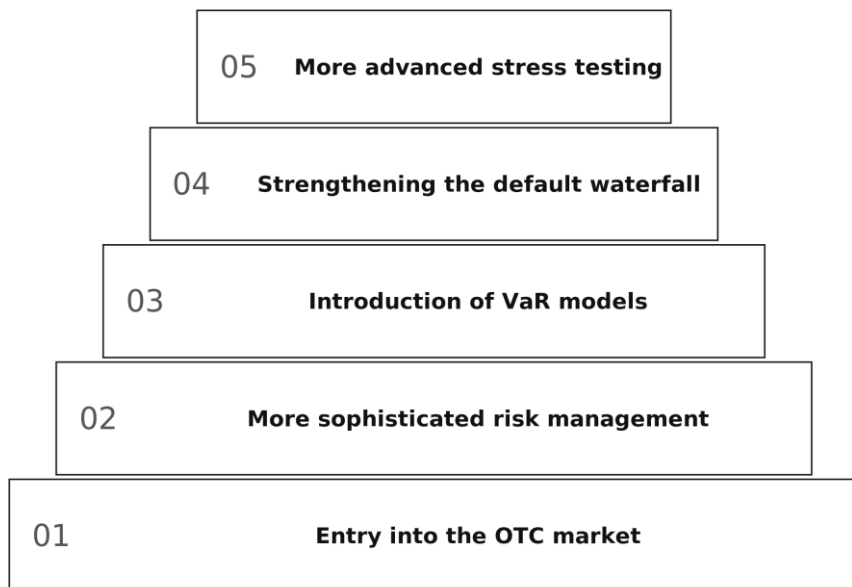
- **Expansion into the OTC market:** The CCP's principal arena expanded greatly, from traditional exchange-traded derivatives into the vast OTC derivatives market (above

all interest rate swaps and CDS indices).

- **More sophisticated risk management:** In taking on the more complex risks of OTC derivatives, CCPs were required to strengthen and upgrade their risk-management frameworks substantially — refining their margin models (introducing VaR-type models and the like), reinforcing their default waterfalls, and enhancing their stress testing. (These risk-management techniques are explained in detail in Chapter 5.)

Clearing was thereby positioned as still more vital infrastructure supporting the stability of the financial system as a whole. The “second phase” in the history of clearing began here in earnest.

The Expanding Role of the CCP



4 Toward the Modern Clearing Regime

Following the international agreements at the G20, jurisdictions put concrete legislation and regulation in place — the Dodd-Frank Act in the United States, EMIR in Europe, and amendments to the Financial Instruments and Exchange Act in Japan (see Chapter 4 for details). CCPs were thereby required to build and operate rigorous risk-management frameworks aligned with international standards (the PFMI), and their functions were strengthened substantially.

JSCC, too, moved with this tide: it expanded into OTC derivatives clearing — beginning to clear yen interest rate swaps in 2012 — and has further deepened its role at the core of Japan's market infrastructure.

Thus, through a history of relentless challenges in risk management, the modern clearing system has evolved into a regime that covers both exchange-traded and OTC derivatives, delivering advanced CCP-centered risk management and efficient operations.

Section 3 The Players in the Clearing Ecosystem: The Roles of CCPs, Brokers, and Clients

As outlined in Section 1.5, the clearing ecosystem rests on multiple players, each with its own roles and responsibilities. This section explains concretely what roles the ecosystem's principal participants — **the CCP and the clearing brokers (clearing participants)**, together with the clients — perform, and how they relate to one another.

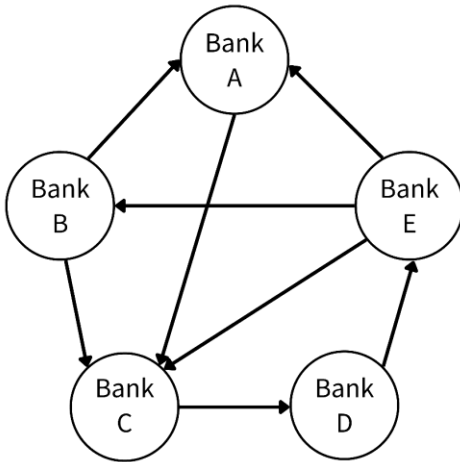
1. The Role of the CCP: The Linchpin of Market Stability

The CCP (central counterparty) is the very “heart” of the clearing system, playing the central role in supporting market-wide stability and efficiency. As we saw in Section 1.2, the CCP interposes itself between the parties to a trade as a “trusted intermediary,” assuming counterparty risk and guaranteeing the performance of settlement.

Core Functions of a CCP

The principal functions through which a CCP contributes to market stability are as follows.

1.1 Centralized Management and Mitigation of Counterparty Risk:



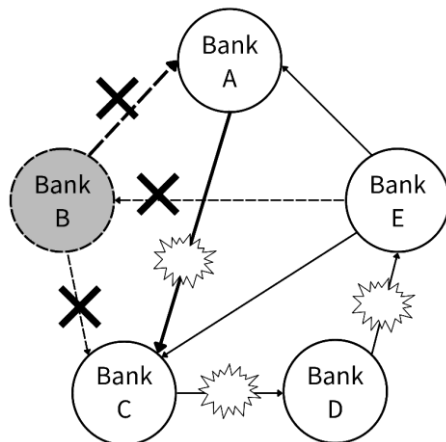
Financial markets rest on a complex web of interconnections among a large number of financial institutions.

In the absence of a CCP, the failure (default) of a single financial institution (e.g., Bank B) can propagate through its trading counterparties in a chain reaction, potentially throwing the entire market into turmoil (systemic risk).

Suppose Bank B defaults. Bank A cannot receive the settlement funds due from Bank B and is therefore unable to pay Bank C; Bank C, in turn, cannot pay Bank D. Bank D has no direct dealings with Bank B, yet because the payment from Bank C fails to arrive, it runs into difficulty meeting its own payment to Bank E. It was precisely this kind of contagion that propagated during the 2008 global financial crisis.

Once a trade is executed, the CCP becomes the seller to the original buyer and the buyer to the original seller (**assumption of obligations, or novation**). As a result, even if an individual counterparty defaults, the risk is concentrated at the CCP, preventing direct contagion to other market participants.

Through this risk-insulation function (**reduced interconnectedness**), market participants can trade with



confidence across a larger number of counterparties without having to assess — or worry about — the creditworthiness of each one individually.

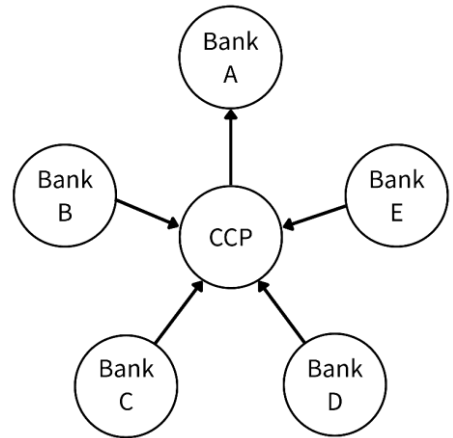
1.2 Guarantee of Settlement Performance:

Backed by a rigorous risk management framework (described below), the CCP guarantees final settlement even when a clearing participant defaults. This preserves confidence in the market as a whole.

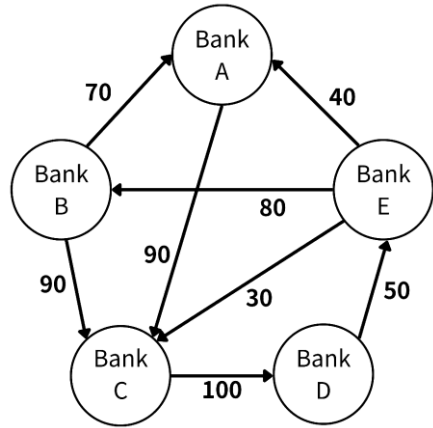
1.3 More Efficient Trade Processing (Multilateral Netting):

One of the most important functions of a CCP is multilateral netting: aggregating the enormous volume of trades among many participants and offsetting the resulting claims and obligations on a multilateral basis.

The case of bilateral trading without a CCP is shown on the right. The arrows indicate the direction of cash flows. For example, Bank A receives 70 from Bank B, pays 90 to Bank C, and receives 40 from Bank E. Here, Bank A exchanges funds directly with Banks B, C, and E.



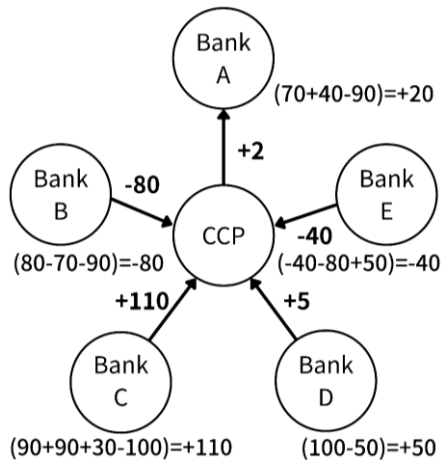
When a CCP is interposed in these trades, all cash flows are consolidated into a single relationship with the CCP. That is, Bank A settles its receipt of 70, payment of 90, and receipt of 40 with the CCP in one combined movement — a net receipt of 20. The figures for the other banks are shown in the diagram below (e.g., Bank B = $-70 + 80 - 50 = -40$). This is referred to as **exposure reduction**.



This is multilateral netting, and it generates several distinct benefits.

1.4 Reduction of Counterparty Risk:

Previously, Bank A traded with Banks B, C, and E and had to monitor the credit standing of each at all times. With its trades now consolidated at the CCP, Bank A no longer needs to concern itself with the creditworthiness of those counterparties.



1.5 Improved Funding Efficiency:

In a bilateral setting, Bank A must constantly track when its receipt of 70 from Bank B, its payment of 90 to Bank C, and its receipt of 40 from Bank E will each fall due. If the receipts from Banks B and E arrive first, Bank A can pay Bank C without setting aside its own cash; depending on the timing of the receipts, however, it may have to fund the payment of 90 to Bank C from its own resources first. With receipts of $70 + 40$ and a payment of 90, Bank A moves a gross total of 200 in funds. Where bilateral trading thus entails gross cash movements of 200, CCP clearing reduces this to a net movement of 20. The resulting transparency and efficiency in funding, in turn, allow cash to be deployed more efficiently.

1.6 Operational Efficiency (Reduced Transaction Processing):

Clearing also delivers substantial efficiencies from an operational standpoint. In bilateral trading, Bank A must make three separate cash settlements — one each with Banks B, C, and E — whereas under CCP clearing only a single settlement with the CCP is required. For a large portfolio spanning multiple products and currencies, the savings are even greater. Suppose, for example, interest rate swap trades in five currencies — U.S. dollars, euros, sterling, yen, and Hong Kong dollars. Bilateral trading would require settlements in five currencies with each of three counterparties, or 15 separate cash movements, whereas CCP clearing requires only five.

1.7 Reduction of Disputes:

In addition, bilateral trading frequently gives rise to disputes (disagreements over margin amounts). Margin calculations for derivatives are complex: calculation logic differs across financial institutions, and discrepancies can arise from differences in calculation timing. When a dispute occurs, the exchange of margin can even grind to a halt. Under CCP clearing, the CCP performs the margin calculation centrally for all

participants, so such disputes do not arise.

1.8 Enhanced Transparency:

- The CCP aggregates trade information and provides transparent information (e.g., its risk management rules and quantitative disclosures) to market participants and regulators. Moreover, rather than maintaining individual contracts with each counterparty, a CCP operates under its own rulebook. In line with CPMI-IOSCO standards, CCPs publish Public Quantitative Disclosures (PQD). All of this strengthens confidence in the market as a whole.
- These disclosures rest on the Principles for Financial Market Infrastructures (PFMI) issued by CPMI-IOSCO, together with the accompanying Disclosure framework and the Public quantitative disclosure standards for CCPs, and are published internationally in a standardized format.

The CCP's Risk Management Framework (Overview)

To support these functions, CCPs build and operate sophisticated risk management frameworks.

- Margin management:** The CCP collects initial margin (IM) and variation margin (VM) from participants to provide against potential losses. (See Section 1.3; calculation models and related topics are covered in Chapters 2 and 5.)
- Default management (the waterfall):** To prepare for a participant default, the CCP maintains a multi-layered loss-absorbing mechanism comprising margin, the default fund (DF), the CCP's own capital (SIG, "skin in the game"), and other resources. (Discussed in detail in Chapter 5.)
- Stress testing:** The CCP regularly runs tests based on extreme market scenarios to verify the adequacy of its financial resources. (Discussed in detail in Chapter 5.)

Through these comprehensive roles and risk management practices, CCPs provide an indispensable foundation for the stability and efficient functioning of financial markets.

2. The Role of the Clearing Broker: Gateway to the CCP, Guardian of the Market

As discussed earlier, the CCP serves as the “heart” of the clearing system, sustaining market stability. Yet the entity that serves as the vital “arteries” connecting that heart to the great majority of market participants (clients) who actually execute derivatives trades — and that stands as a “guardian” on the front line of risk management — is the **clearing broker (also known as a clearing participant or FCM)**.

2.1 Direct Clearing vs. Client Clearing

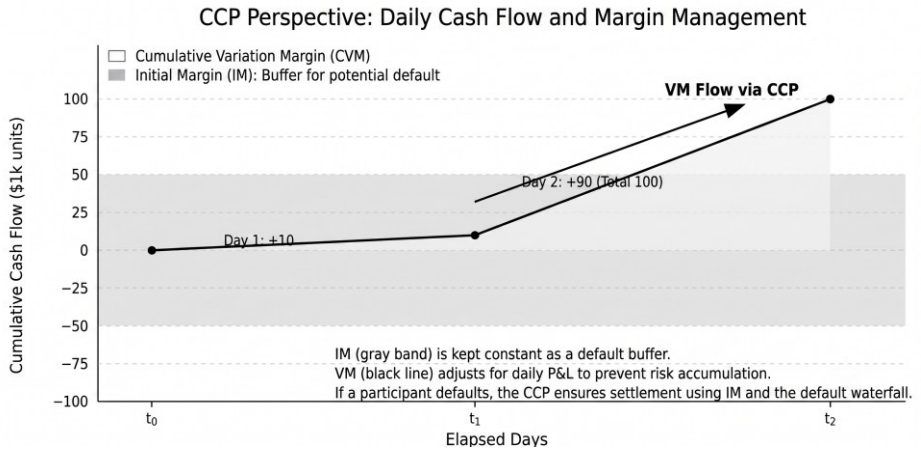
There are two principal ways to access clearing services.

- **Direct clearing:** Financial institutions that are clearing members of the CCP (chiefly major banks and securities firms) clear **their own trades** directly at the CCP. Stringent membership requirements apply.
- **Client clearing:** Clients that are **not** clearing members of the CCP (asset managers, hedge funds, pension funds, insurance companies, non-financial corporates, and others) access the CCP’s clearing services indirectly, **through a clearing broker**.

In client clearing — the subject of this book — the clearing broker plays an indispensable role. That role extends far beyond mere intermediation, encompassing risk management, operations, compliance, and client protection. Having spent many years of my career in this business, I devote this section to a detailed examination of the clearing broker’s role.

2.2 Specific Roles and Functions of the Clearing Broker

The principal functions a clearing broker provides to its clients and to the market as a whole can be summarized as follows.



2.3 Providing Access to CCPs (Access Services)

- Because the clearing broker satisfies the membership requirements imposed by many CCPs — stringent financial standards and operational capabilities — clients can use the CCP's clearing infrastructure without meeting those requirements directly.
- By providing connectivity to multiple CCPs (e.g., JSCC, LCH, and CME) and exchanges, the clearing broker gives clients access to global markets through a single point of contact.

2.4 Margin and Collateral Management and Execution of Clearing Operations (The Core of Risk Management Services)

- Margin calculation, calls, and deposit: Based on the initial margin (IM) and variation margin (VM) requirements set by the CCP, the broker calculates, collects, and deposits clients' margin with the CCP. It reliably executes the daily exchange of VM arising from mark-to-market valuation.

- Collateral management: The broker values and administers the collateral deposited by clients (cash, government bonds, equities, and so on). It assesses collateral eligibility, applies haircuts, monitors collateral values, and issues calls for additional collateral (margin calls) as needed.
- Collateral optimization and transformation: To enhance clients' capital efficiency, the broker may also offer services that optimize deposited collateral or transform it into collateral eligible for acceptance by the CCP (e.g., converting non-cash collateral into cash collateral). (See Chapter 5 for details.)

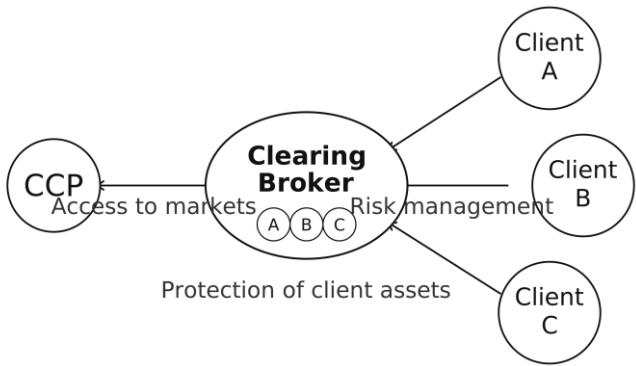
2.5 Risk Management and Credit Functions

- Credit assessment and credit limit setting: The broker evaluates each client's creditworthiness (financial condition, trading history, and so forth), assesses the potential risks associated with the client's trading, and sets and manages appropriate credit limits (portfolio limits, intraday trading limits, and the like).
- Guarantee to the CCP: The broker guarantees the client's performance of its trades to the CCP. In effect, the clearing broker assumes the client's default risk vis-à-vis the CCP.
- Default management: Should a client fail to meet a margin payment (default), the clearing broker first advances the payment to the CCP from its own capital and other resources. At the same time, it takes measures to contain the risk, such as compulsorily liquidating (closing out) the client's positions. (See Chapter 5 for details.)

2.6 Protection of Client Assets

- This is one of the clearing broker's most important responsibilities. In accordance with regulation (e.g., the CFTC rules in the United States and EMIR in Europe), margin and collateral deposited by clients are **held in clear legal segregation** from the clearing broker's own assets.
- As a result, even if the clearing broker itself fails, client assets are more likely to

be insulated from the broker’s bankruptcy estate and protected (**bankruptcy remoteness**). The specific segregation models (LSOC, ISA, and others) are described in detail in Chapter 6.



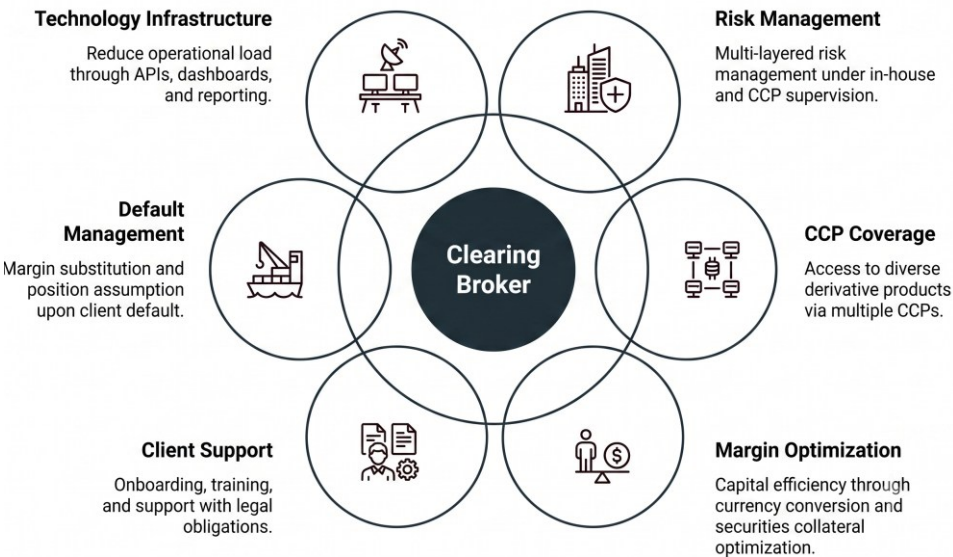
2.7 Functions That Support the Stability of Market Infrastructure

The clearing broker is not a mere intermediary: it performs systemically important functions that underpin the stability of the financial market infrastructure as a whole. The framework set out by the FIA (Futures Industry Association) highlights the following dimensions.

Function	Description
Multi-layered risk management	In addition to its own credit risk management (e.g., credit limits and collateral management), the broker is embedded in the oversight frameworks of CCPs and regulators, forming a three-tier line of defense.
Multi-CCP and product coverage	Provides access across derivatives products (IRS, CDS, futures, and others) through connectivity to multiple CCPs (JSCC, LCH, CME, and others).
Margin and collateral optimization	Optimizes currency conversion and the valuation of securities collateral to improve the capital efficiency of margin requirements. Includes support for posting collateral in the form the CCP requires.
Client support and education	Provides training and documentation covering onboarding paperwork, explanation of collateral requirements, compliance with legal obligations, and related matters.
Default management	Upon a client’s failure, pays margin to the CCP on the client’s behalf, takes over positions, and cooperates in the auction process, among

support	other measures.
Technology platform and operational support	Reduces clients’ operational burden by providing APIs, risk management dashboards, statements, and a range of reports.

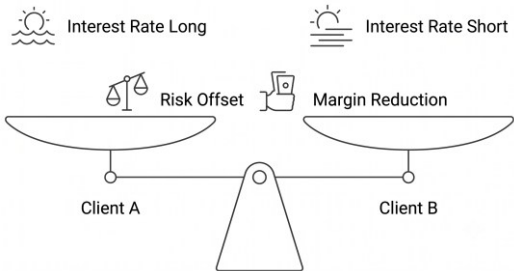
Functions of a Clearing Broker



2.8 Risk Diversification through Consolidation of Multiple Client Portfolios:

- By aggregating trades from a large number of clients, the clearing broker performs the distinctive function of **optimizing risk across its portfolio as a whole.**

•**The principle of portfolio offset:** Where one client’s positions and another’s run in opposite directions in risk terms (e.g., long versus short exposure to rising interest rates), the risks offset one another across the broker’s overall portfolio, potentially reducing the margin — in particular the IM — that the broker must post to the CCP.



•**Economic significance:** This diversification benefit is one that individual clients could not achieve by facing the CCP directly, and it constitutes one of the clearing broker’s key sources of economic value. It improves the broker’s own capital efficiency, which in turn can translate into lower commissions and enhanced services for clients. (Taken up further in the discussion of risk management in Chapter 5.)

Item	Description
The principle of portfolio offset	Where Client A’s positions and Client B’s positions run in opposite directions in risk terms (e.g., one long and the other short against rising interest rates), the risks offset one another across the overall portfolio , reducing the total margin (IM/VM) required of the CB. (Arises mainly in futures omnibus accounts.)
Collateral negotiating power with the CCP	A consolidated portfolio allows the broker to present more refined risk assessment models and collateral optimization strategies to the CCP. This makes it possible to exploit multilateral netting and cross-margining frameworks.
Cost optimization	Lower margin requirements reduce costs for clients while also improving the CB’s own capital efficiency.
Ex-post monitoring and stress response	On the consolidated portfolio, concentration risk analysis and stress testing are carried out to keep the overall risk structure optimal.

2.9 Operations, Reporting, and Compliance Support:

◦The broker carries out the complex day-to-day **operational tasks** of trade

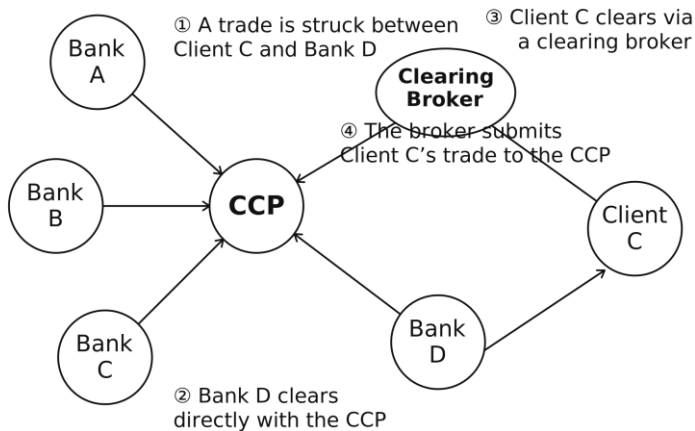
affirmation, submission of trade data to the CCP, settlement instructions, position management, and the preparation and delivery of various reports (trade statements, margin statements, and the like).

- It discharges **trade reporting obligations** to regulators (e.g., trade repository reporting under EMIR) on clients' behalf, or supports clients in fulfilling them.
- It provides information on increasingly complex domestic and international regulatory developments and supports clients in **building their compliance frameworks**.

3. How Client Clearing Works: The Difference Between the Trading Counterparty and the Clearing Counterparty

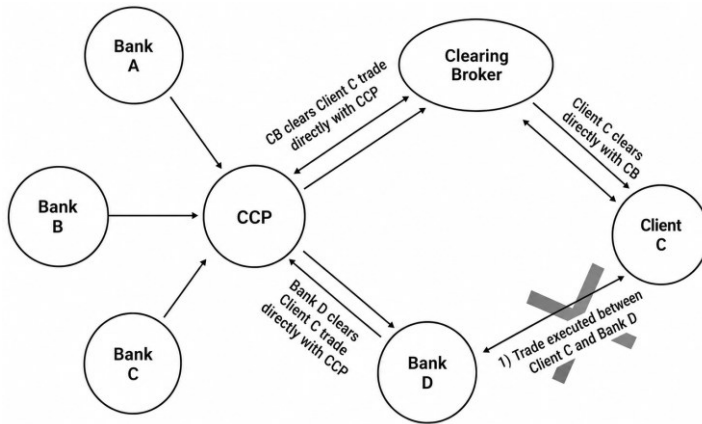
Client clearing is sometimes the subject of a particular misunderstanding.

At first glance, this diagram appears to suggest that the client can trade only with its clearing broker. The diagram, however, depicts credit relationships. In practice, as shown here, the client can trade with any counterparty.



Once CCP clearing has been completed, the flow of the trade changes as follows.

The trading relationship between Client C and Bank D is converted into trades through the CCP: Bank D now clears directly with the CCP, while Client C clears through its clearing broker (CB).



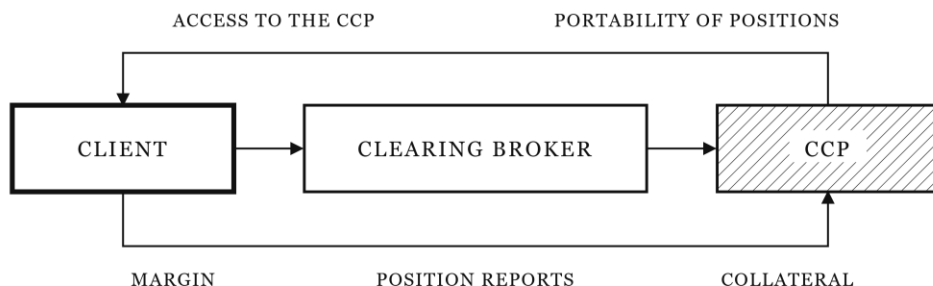
The trading relationship between Client C and Bank D is thereby severed and decomposed into (1) a trade between Bank D and the CCP and (2) a trade between Client C and the CB (via the CCP). In addition, just as in direct clearing, the client can achieve multilateral netting across multiple counterparties through its clearing broker. The client thus enjoys the CCP's guarantee and the benefits of netting through its clearing broker, while the clearing broker manages the risks of multiple clients and bears responsibility to the CCP.

The Importance of the Clearing Broker

Clearing brokers are far more than trade intermediaries. By providing access to CCPs, executing complex operations, protecting client assets, and — above all — standing on the **front line of risk management**, they are indispensable to the stability and efficiency of the entire clearing ecosystem. Their role is becoming ever more sophisticated and specialized as regulation tightens and technology advances. (The future of the clearing broker is discussed in detail in Chapter 7.)

4. The Role of the Client: As a User of Clearing Services

The ultimate beneficiaries of the clearing ecosystem — and the parties who actually enter into derivatives trades — are the clients (customers, or end users). Asset managers, hedge funds, pension funds, insurance companies, non-financial corporates, and others vary widely in their profiles and trading objectives (differences by client type are detailed in Chapter 6, Section 6.2), but in using clearing services they



all assume the following common roles and responsibilities.

1. Trade Execution and Delegation of Clearing:

- The client **executes** derivatives trades on exchanges, on electronic trading platforms, or bilaterally with dealers.
- For each executed trade, the client **delegates clearing** to its designated clearing broker. This delegation process and the related agreements (such as the CDEA) are covered in Chapter 6.

2. Selecting and Contracting with a Clearing Broker:

- The client **selects a clearing broker** suited to its needs (products traded, service levels, costs, and so on) and enters into a clearing agreement, collateral

agreements, and related contracts. Evaluating the broker's creditworthiness and risk management framework is likewise important. The selection process and the contractual documentation are explained in detail in Chapter 6, "Onboarding and Documentation."

3. Provision of Margin and Collateral:

- Through its clearing broker, the client is obligated to **provide, without delay, the initial margin (IM) and variation margin (VM)** required by the CCP and the broker.
- The client must hold and manage assets eligible as collateral (cash, government bonds, and other high-quality liquid assets, or HQLA). Margin calculation methodologies and the details of collateral management are explored in depth in Chapter 2, Section 2.3, Chapter 3, Section 3.2, and above all in Chapter 5, "Risk Management in Clearing."

4. Risk Management:

- While clearing greatly reduces counterparty risk, market risk, liquidity risk, and other risks remain with the client. Each client must **build and operate an appropriate risk management framework** consistent with its own risk tolerance.
- The client must also observe the trading limits and other parameters set by its clearing broker. The importance of the client's own risk management framework is taken up again in **Chapter 5**.

5. Regulatory Compliance and Reporting:

- The client must comply with applicable financial regulations (e.g., trade reporting obligations) and provide the necessary information to its clearing broker and to regulators. The relevant legal and regulatory landscape is outlined in Chapter 4, "The Legal and Regulatory Framework for Clearing and Regional Characteristics."

- The client is also expected to understand and abide by the rules established by the CCP and the clearing broker.

When clients properly discharge these roles and responsibilities, they contribute to the smooth operation and stability of the clearing system as a whole — and, ultimately, to the protection of their own interests. For readers actively considering the adoption of clearing, the detailed treatment of the onboarding process and documentation in **Chapter 6** should prove a substantial practical aid.

Section 4: Market Size and Significance

The clearing market functions as the “safety net” of the financial system, underpinning the stability and efficiency of trading. This section examines the size of the global clearing market and the role it plays in financial markets overall, covering both over-the-counter (OTC) derivatives and exchange-traded derivatives (ETD). We pay particular attention to the importance of how the buy side (investors) makes use of clearing.

1. The Size of the Global Clearing Market

The clearing market grew rapidly in the wake of the 2008 global financial crisis, driven by tighter regulation and heightened risk management needs; today it is an indispensable foundation of the international financial markets. This section unpacks the overall picture through concrete figures and examples.

1.1 The Overall Market: Size and Growth

●The size of the over-the-counter (OTC) derivatives market

According to Bank for International Settlements (BIS) data for end-December 2025, the total notional amount of OTC derivatives outstanding is approximately USD 845 trillion. This figure aggregates the notional amount of contracts entered into bilaterally by financial institutions and includes interest rate swaps, credit default swaps (CDS), and currency swaps. As of the same date, interest rate swaps alone accounted for roughly 65% of all OTC notional outstanding (about USD 545 trillion), and CCP clearing has become the linchpin of risk mitigation.

Example: An interest rate swap is used much like “insurance that locks in future interest rates,” and such swaps account for roughly 70% of OTC market volume.

●The size of the exchange-traded derivatives (ETD) market

ETDs such as futures and options, by contrast, are traded as standardized contracts on exchanges. According to the Futures Industry Association (FIA), global ETD

volume in 2023 reached roughly 15 billion contracts, with aggregate notional on the order of several hundred trillion dollars. These contracts are cleared automatically through an integrated exchange-and-CCP framework and fall outside the BIS OTC statistics.

Example: Nikkei 225 futures give investors a tool for taking exposure to, or hedging against, broad market movements, and their clearing is handled by JSCC.

●Drivers of market growth

A major driving force was the G20 commitment, made in the wake of the 2008 global financial crisis, to mandate central clearing of OTC derivatives. The role of CCPs expanded accordingly, the aim being to reduce counterparty risk — the risk that the other side of a trade fails to pay. For interest rate swaps and CDS in particular, CCP clearing became all but mandatory as a matter of regulatory compliance. ETDs, by contrast, had clearing built into their design from the outset, and their growth has been correspondingly stable.

2.2 Characteristics of Regional Markets

- Europe:** LCH (the London Clearing House) is the dominant player, clearing roughly 86% of centrally cleared euro-denominated interest rate swaps, forward rate agreements included (2025 Q3), with transparency and risk management rigorously enforced under the EMIR regime.

Example: the clearing of euro-denominated swaps could be likened to a pillar supporting the very “heart of European finance.”

- United States:** CME holds the leading share in futures and options, while ICE is strong in energy and CDS. The Dodd-Frank Act strengthened the clearing of OTC derivatives, enhancing the safety of the market.

Example: CME’s interest rate swap clearing might be called “a shield for U.S. financial stability.”

- Asia:** In Japan, JSCC covers a broad range of products, from equities to commodity

futures. China and India are also building their own CCPs. Going forward, international interoperability will be key.

Example: JSCC plays a role akin to tying Japan's markets together with "a single thread."

2.3 OTC and ETD at a Glance

Item	Over-the-counter (OTC) derivatives	Exchange-traded derivatives (ETD)
Form of trading	Bilateral contracts (customizable)	Standardized on exchanges
Representative examples	Interest rate swaps, CDS, currency swaps	Equity index futures, government bond futures, options
Clearing method	CCP clearing mandate (for in-scope products)	Automatic clearing via exchange plus CCP
Statistics	Compiled by the BIS	Tracked separately by the FIA and exchanges
Image	"A tailor-made suit"	"An off-the-rack suit"

Note: in the future, growing interconnection among global CCPs may blur the boundary between OTC and ETD.

2 The Role of the Clearing Market

●The linchpin of risk management

A CCP acts as a "shock absorber" that keeps the market from collapsing even when a counterparty fails. During the COVID-19 shock of 2020, CCPs kept clearing stable and contained systemic risk.

●Importance to the buy side

Buy-side participants such as hedge funds and pension funds rely on clearing to secure both safety and cost efficiency in their trading. Particularly since the post-crisis tightening of regulation, trading through CCPs has become essential, and the barriers to market participation have come down.

●Challenge: As reliance on CCPs deepens, "concentration risk" — the risk that a CCP

itself fails — has come under scrutiny, and oversight has been strengthened under international standards (CPMI-IOSCO).

The clearing market comprises OTC derivatives (approximately USD 845 trillion at end-December 2025) and ETDs (roughly USD 107 trillion of open interest on the BIS measure, which covers interest rate and foreign exchange futures and options only), each supported by distinct needs and infrastructure. As the BIS and FIA statistics indicate, these markets can be expected to keep growing and evolving as the foundation for risk management and greater transparency.

Column A: Trading and Clearing at the Eighteenth-Century Dojima Rice Exchange

Introduction

The time is the eighteenth century; the place, Osaka, Japan. Here stood a market still internationally renowned today as one of the world’s earliest organized commodity futures exchanges: the Dojima Rice Exchange. This market, which has drawn the attention of scholars including Lambert (2011) and Takatsuki (2018), played a central role in the economy of early modern Japan. In this column, drawing on recent research, we unfold the story of how this pioneering market functioned — its trading and clearing arrangements, and the institutions that supported them.

The Founding of the Dojima Rice Exchange — A Market Born of the “Nation’s Kitchen”

During the Edo period, Osaka established itself as the “kitchen of the nation” — the hub to which annual tax rice from across Japan was shipped and sold. Initially, rice was traded at the storefronts of powerful merchants, most famously the House of Yodoya. By the latter half of the seventeenth century, however, the feudal domains were building warehouse-residences (*kurayashiki*) one after another in the vicinity of Dojima and Nakanoshima. These warehouses issued “rice bills” (*kome kitte*) certifying ownership of the rice they stored, and the circulation of these bills grew increasingly active (Takatsuki 2012).

Intriguingly, these rice bills came to represent not only rice physically held in the warehouses (*shomai*) but also rice scheduled to arrive in the future (*kaimai*, rice in transit), and even claims on rice that had yet to be harvested. As a result, rice bills themselves became objects of speculative trading (Takatsuki 2018).

Confronted with this surge of trading and speculation, the Tokugawa shogunate finally acted to stabilize rice prices and preserve order in the market. In 1730 (Kyoho 15), the shogunate officially recognized the rice market at Dojima, and the Dojima

Rice Exchange was established. This put in place an organization and institutions recognizable in modern exchanges — a membership system, explicit trading rules, and, as we shall see in detail below, a clearing function (Takatsuki 2012; JPX materials). The rice prices set at Dojima became the benchmark for rice quotations throughout the country, exerting a powerful influence on the Japanese economy as a whole (Takatsuki 2018).

Trading at the Dojima Rice Exchange: From Physical Rice to Promises of the Future

Two main types of trading were officially sanctioned at the Dojima Rice Exchange.

1. **Shomai trading (shomai akinai):** This was the buying and selling of the rice bills themselves, as issued by the warehouse-residences. In substance, it was a market for trading physical rice — or, more precisely, the right to exchange a bill for it (Takatsuki 2012).
2. **Choaimai trading (choaimai akinai):** This was a futures transaction referencing a standard grade of rice and specified future delivery dates (normally three each year). Trades were recorded on the books, and no physical rice actually changed hands. By convention, choaimai trading took place ahead of shomai trading, and its prices were regarded as a leading indicator of movements in physical rice prices (Takatsuki 2012; Takatsuki 2018). Most speculative trading was conducted in this choaimai market.

The “rice bills” and similar rights that were traded in fact came in several variations. The shogunate repeatedly attempted to restrict speculative “empty rice” trading — trading not backed by physical rice — but in practice a wide range of such rights were actively traded in the market (Takatsuki 2012).

- **Dekitte (physical rice bills):** Rice bills certifying that physical rice was held in a warehouse-residence; these were traded mainly in the shomai market.

●**Claims on future rice (the subject of choaimai trading):** In choaimai trading, what changed hands was the right to receive, or to deliver, rice at a specified future time. This included rights to rice still in transit (*kaimai*) and to rice not yet harvested, giving the trades, in substance, the character of “empty rice” transactions (Takatsuki 2012).

The standard trading unit was ten *koku* of rice. In *shomai* trading, prices were set by auction among the rice brokers, with the rice bill passing to the winning bidder. In choaimai trading, prices were formed through open-outcry auction (*seri baibai*) conducted at the exchange (Takatsuki 2012).

How, then, was settlement effected? *Shomai* trading was settled, as a rule, by physical delivery — the rice bill exchanged against payment. In choaimai trading, a more sophisticated method prevailed: by entering into an offsetting transaction before the delivery date (selling back a contract one had bought, or buying back a contract one had sold), participants exchanged only the difference in value. This “**cash settlement**” was widely practiced (Takatsuki 2018). Contracts held to the delivery date were ultimately settled either by delivery of rice bills or by payment of money in lieu.

Choaimai trading was not merely a venue for speculation. It was also skillfully used as a means of hedging against price fluctuations. A daimyo or merchant shipping rice from the provinces to Osaka, for example, feared the risk that rice prices would fall while the cargo was in transit. He would therefore sell futures (*choaimai*) at Dojima in advance, in an amount matching the rice being shipped — a practice known as “**hedge selling**” (***uritsunagi***) (Takatsuki and Kamihigashi 2022). If the price of rice had indeed fallen by the time the cargo reached Osaka, he could buy back the futures contracts at the lower price, realizing a gain that offset the loss on the physical rice.

A Clearing System Ahead of Its Time: Safeguarding the Integrity of

Trades

Behind the remarkable vitality of choaimai trading lay an elaborate clearing mechanism that secured the safety of transactions.

Within the Dojima Rice Exchange stood a dedicated clearing facility known as the “**keshiaiba**”. Here, the

settlement of trades — above all the cash settlements so frequent in choaimai trading — was processed on a centralized basis (Takatsuki 2007; Takatsuki 2012).

The complex settlement operations of the keshiaiba were handled, in their entirety, by a group of powerful money changers known as the “**yarikuri**

ryogae”. Far from performing mere settlement administration, they interposed themselves as intermediaries between individual buyers and sellers. Crucially, if one party became unable to pay (a default), they guaranteed payment to the other party. This was a groundbreaking arrangement that can be regarded as a prototype of the **clearing members** and the **novation** function of a modern central counterparty (CCP) (Takatsuki 2007; Takatsuki 2012). It is fair to say that the presence of the yarikuri ryogae insulated participants from the credit risk of individual counterparties and dramatically enhanced confidence in the market as a whole.

What happened, then, if a market participant — or indeed a yarikuri ryogae itself — fell into settlement default (“dishonor”)? In such cases, the shogunate could ultimately intervene. To prevent market disruption, it sometimes purchased the dishonored rice bills or claims with public funds, or compulsorily liquidated the



assets of the insolvent party (shindai kagiri) (Takatsuki 2012). The shogunate thus served as the ultimate backstop — a safety net — for maintaining confidence in the market.

No Margin System? Dojima's Diverse Approaches to Risk Management

In modern futures trading, margin requirements and mark-to-market — the practice of revaluing positions at current prices and calling additional margin as losses accrue — are indispensable to the safety of trading. Remarkably, however, current research has not confirmed that either existed in any explicit form at the Dojima Rice Exchange (see Takatsuki 2012).

This does not mean, however, that Dojima lacked risk management. On the contrary, credit risk was managed adroitly through a diverse, multilayered set of arrangements suited to the social conditions of the time (Takatsuki 2007; Takatsuki 2012; Takatsuki 2018).

- **Strict admission requirements:** Becoming a rice broker or a *yarikuri ryogae* required passing a rigorous screening of one's assets, identity, and standing.
- **Mutual monitoring through guild organizations:** Brokers and money changers formed guilds and kept one another's financial condition under constant watch. If a member appeared to be in difficulty, measures such as restricting that member's trading could be taken.
- **Limited mutual liability:** It has also been suggested that arrangements may have existed under which, if a member of the guild defaulted, the other members would absorb a portion of the losses.
- **Shogunal intervention:** The shogunate maintained market discipline by imposing severe punishments — banishment, confiscation of assets — for misconduct or insolvency.

Operating in combination, these mechanisms are thought to have preserved

confidence in the market even without a quantitative margin system in the modern sense.

The Shogunate and the Market: Oversight, Intervention, and the Legal Foundation

The Tokugawa shogunate officially sanctioned the establishment of the Dojima Rice Exchange and supervised its operation, while also involving itself deeply in the workings of the market.

- **Market oversight and regulation:** The shogunate took part in setting trading rules and appointing officers, seeking to maintain order in the market. In particular, when it judged speculation to have gone too far, it sometimes temporarily banned choaimai (“empty rice”) trading or restricted trading periods. Yet because it also understood the importance of market vitality and of the hedging function, outright prohibition was rare (Takatsuki 2012).
- **Market intervention:** When rice prices spiked in times of famine or, conversely, collapsed, the shogunate intervened to stabilize the market. Specifically, it restricted or encouraged kaimai (shipments of rice from the provinces to Osaka), bought and sold rice from its own holdings in the market, and at times even ordered daimyo and merchants to buy or sell rice (as in the orders to sell off long rice positions) (Ito 1993; Takatsuki 2018).
- **Legal framework and dispute resolution:** The shogunate protected rice bills as property rights and laid down certain legal rules governing the claims and obligations arising from choaimai trading. When disputes arose over transactions, the parties could bring suit before the shogunate’s judicial organs (such as the Osaka town magistrate’s office) to seek resolution. The research of Yasuo Takatsuki has shown in detail how this shogunal judicial system shaped the development of the Dojima market (Takatsuki 2012; Takatsuki 2018).

The Dojima Rice Exchange as a Mirror of Its Age

Eighteenth-century Japan saw the dramatic development of a nationwide market economy, with the monetary economy — centered above all on Osaka — penetrating every corner of society. Within it, rice was not merely food: it underpinned the fiscal foundations of the feudal domains and served as a measure of value, playing a role akin to that of money itself (Takatsuki 2018). Most domains shipped the rice they collected as annual tax to Osaka, converted it into cash there, and ran their finances on the proceeds. Movements in rice prices therefore had a direct impact on the management of the domains and, by extension, on the economic fortunes of the entire samurai class (Takatsuki 2012). Against this backdrop, the Dojima Rice Exchange — the benchmark for rice prices nationwide, a stabilizer of prices, and the backbone of efficient rice distribution — was indispensable to the Japanese economy as a whole.

Conclusion: An Edo-Period Market System of World Renown

The eighteenth-century Dojima Rice Exchange was a highly sophisticated market, combining physical trading based on rice bills (shomai trading) with an advanced form of futures trading (choaimai trading). It featured a centralized clearing function operated through the keshiaiba and the yarikuri ryogae, and it combined standardized contracts (choaimai), efficient



mechanisms for trade execution, centralized management of credit risk by a clearing facility, and the legal and institutional backing of the shogunate. The exchange thus already embodied many of the elements found in modern exchange systems, and it

can fairly be said to have realized a market system that was exceptionally advanced for its time anywhere in the world (Takatsuki 2012; Lambert 2011).

It is true that the modern risk management tools of margin and mark-to-market are nowhere to be found. Yet confidence in the market was firmly maintained through a diverse array of qualitative and institutional safeguards: restricted membership, mutual monitoring by the guilds, and shogunal intervention. The Dojima Rice Exchange not only contributed to the economic development of early modern Japan; its innovative design also shines as a remarkable chapter in the history of the world's derivatives markets.

Glossary

- **Rice bills (kome kitte):** Certificates of rice storage and exchange issued by the domains' warehouse-residences. They could represent not only physical rice but also claims on future rice.
- **Shomai trading:** The buying and selling of physical rice bills; the equivalent of a spot market.
- **Choaimai trading:** Futures trading in which contracts were made to buy or sell standard rice at a predetermined price at a fixed future date. Cash settlement was the predominant method.
- **Cash settlement:** A settlement method in which, through an offsetting transaction, only the difference between the original contract price and the settlement price changes hands.
- **Hedge selling (uritsunagi):** A form of hedging in which a futures position is established opposite to a physical position to mitigate price risk.
- **Keshiaiba:** The clearing facility established within the Dojima Rice Exchange, which centrally processed the settlement of choaimai trades and related business.
- **Yarikuri ryogae:** The powerful money changers who handled settlement at the keshiaiba. By standing as counterparty to trades and assuming credit risk, they

played a role close to that of modern clearing members.

- **Kaimai:** The shipment by sea of annual tax rice from the domains to consumption centers such as Edo and Osaka.

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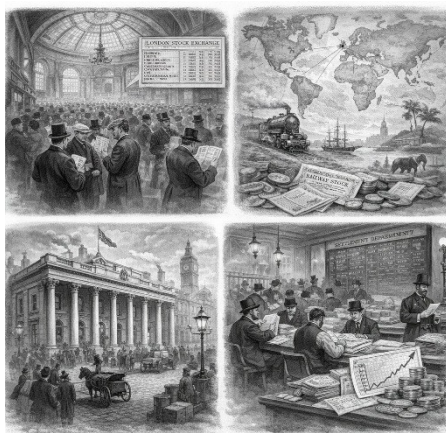
Column B: The Heart of a Financial Empire: The Nineteenth-Century London Stock Exchange and the Dawn of Clearing

Introduction: Tracing the Origins of Modern Financial Infrastructure

Today, the London Stock Exchange Group (LSEG) is a leader in global financial market infrastructure, counting among its subsidiaries LCH, a giant among central counterparties (CCPs). This vast apparatus safely processes an astronomical volume of financial transactions every day and underpins the stability of global capital markets. Trace its roots, and one arrives at the London Stock Exchange (LSE) of the nineteenth century, which rose to the summit of world finance against the backdrop of the Industrial Revolution and the glory of the British Empire. In this column, we unfold the early story of how the LSE, at the height of its prosperity, protected the safety of trading and laid the foundations of clearing and risk management that lead to the modern CCP.

Becoming the Exchange Where the World's Capital Gathered: The Founding and Remarkable Rise of the LSE

To facilitate the smooth trading of the ever-growing stock of government securities, such as British government bonds, and of private company shares, the LSE began its history in 1801 as a formal membership organization [16, Michie 1999]. From its founding, the LSE swiftly ascended to a central position in the capital markets of Britain and the world. Throughout the nineteenth century, shares and bonds were listed one after another — in the new industries of



railways, mining, and the telegraph, and in enterprises connected with the British Empire's colonies across the globe — drawing investors not only from Britain but from around the world to the London market. On the eve of the First World War, it is said that roughly one-third of all the shares available to investors worldwide were traded on the LSE; such was the breadth of its influence. London had become, quite literally, “banker to the world,” and the LSE beat at the very heart of that financial empire.

The Shadow Cast by Expanding Trading: Credit Risk and the Germination of Clearing

As trading volumes exploded, however, a serious problem came to the fore: “counterparty risk” (credit risk) — the risk that the other side of a trade would fail to pay as promised, or fail to deliver securities, on the settlement date. At the LSE in those days, trades were conducted directly between member brokers (who intermediated client orders) and jobbers (specialists dealing for their own account), and settlement ordinarily took place in bulk on “Account Days” set at two-week intervals. If a counterparty failed before the Account Day arrived, payments could be held up in a chain reaction — the market lived with the constant danger of seizing up altogether.

To soften this risk, the LSE established an internal Settlement Department to make processing on Account Days more efficient. There, the trade records of the members were matched against one another, and the net balances of purchases and sales over the settlement period were computed (netting), reducing the amounts of cash and securities that actually had to change hands. This is one of the basic functions of modern clearing — but by itself it could not eliminate the risk of a counterparty's failure.

The Road to Central Clearing and Risk Management: The

Development of Collateral Arrangements

In search of a more fundamental remedy, from the late nineteenth century the LSE began exploring arrangements approaching the functions of a modern central counterparty (CCP). Proposals were debated and attempted, including the establishment of funds through which members would mutually guarantee one another's payments, and ways for the exchange itself to take a more active role in the performance of settlement. But a complete central clearing function — the exchange becoming party to every trade and assuming its obligations (novation) — would have to await a somewhat later era, with the founding of the organizations that became the forerunners of LCH.

What did develop substantially in this period, by contrast, was a system of “collateral” to cover credit risk in practical terms. The LSE came to require trading participants to post collateral (known at the time as “Deposit” or “Cover”) against the potential losses that could arise from the positions they carried [*18, Morgan & Thomas 1962].

- **The role of collateral (margin):** This can fairly be described as the forerunner of the modern “margin” system. Trading participants were required to deposit, as collateral, cash or eligible securities — most notably Consols (British perpetual government bonds), then the most creditworthy instrument available — in amounts commensurate with the risk of their outstanding open positions [*17, Michie 1999]. In the event that a participant failed to settle, the exchange and the other members could draw on this deposited collateral to cover part or all of the resulting losses.

- **Risk-based management:** The amount of collateral required was not uniform. It was customarily adjusted flexibly according to the type of transaction, the riskiness of the securities concerned (such as the volatility of their prices), and — above all — the creditworthiness of the trading participant itself (its reputation, financial condition, and so forth) [*16, Morgan & Thomas 1962]. This reflects the basic

philosophy of risk-based management: demanding more substantial safeguards for riskier transactions and riskier counterparties.

●**Maintaining the value of collateral:** The market value of deposited collateral, particularly securities, fluctuates from day to day. Mechanisms existed under which a participant could be required to post additional collateral if the value of the collateral declined and became insufficient to cover the risk of the position [*16, cf. Michie 1999]. While this differs from daily mark-to-market in the strict modern sense, it was an important step toward keeping the value of collateral consistently adequate and managing risk accordingly.

These collateral and margin arrangements made each trading participant conscious of its own risk and helped restrain reckless speculation. At the same time, they played a decisive role in preventing losses from cascading through the market when a failure did occur, and in strengthening confidence in the market as a whole.

The LSE Innovations That Laid the Foundations of the Modern CCP

In the nineteenth century, the London Stock Exchange enjoyed unprecedented prosperity as the center of the world economy, yet it also faced an ever-present vulnerability: the credit risk generated by the market's rapid expansion. In response, the LSE sought to protect the stability of, and confidence in, the market by exploring centralized clearing and settlement arrangements and, in particular, by introducing and developing risk-based collateral and margin systems. These innovations were extraordinarily advanced for their time, and they can fairly be said to have laid important foundations for the risk management functions — margining, default management, and the like — of the modern central counterparty (CCP).

Modern CCPs, LCH among them, are equipped with far more sophisticated and complex risk management systems, thanks to remarkable advances in technology and the strengthening of regulation. Yet the core philosophy running beneath them — managing counterparty risk and preserving confidence in the market — rests on the wisdom and experience of the pioneers who worked through trial and error amid the bustle of nineteenth-century London. Tracing the history of the LSE is more than an exercise in learning about the past: it offers valuable insight into how the financial system we use today took shape, and how it should evolve into the future.

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Column C: A Pioneer of the Futures Markets: The

CME and Its Trail of Clearing Innovation

Introduction: The Driving Force Behind the World's Largest Derivatives Exchange

Today, CME Group is known as the world's largest and most diverse derivatives exchange group. Interest rates, equity indices, foreign exchange, energy, agricultural products, metals, and even crypto assets — it provides venues for managing the risks of virtually every asset class and for discovering fair prices. Its clearing arm, CME Clearing — the very heart of the group — delivers indispensable stability to the global financial system through ironclad risk management and relentless technological innovation [*29, Risk.net 2023]. Yet the story of this giant group begins in the mid-nineteenth century, in Chicago, with an exchange for the most everyday of agricultural products: butter and eggs. In this column, we trace the exciting trajectory by which the CME leapt from a modest commodity exchange to a leader of the world's financial markets, driving revolution after revolution in the field of clearing along the way.

The Early Years: From Agricultural Markets to the Establishment of a Clearing Function

The CME's direct roots lie in the Chicago Butter and Egg Board, established in 1898 (a separate organization from the Chicago Board of Trade (CBOT), founded in 1848) [*29, CME Group History]. At the outset it was, as its name suggests, a market for trading physical butter, eggs, and similar produce. Before long, however, in response to producers and distributors seeking protection against the risk of future price movements, forward and futures contracts promising future delivery emerged, and trading gradually gathered momentum.

As trading grew more active, however, “counterparty risk” — the risk that a party would not, or could not, honor the contracts it had entered into — became a serious

problem. To confront this challenge head-on, the exchange made a momentous decision in 1919: it established a dedicated clearing house and, at the same time, renamed itself the Chicago Mercantile Exchange (CME). This clearing house was a forerunner of the modern CCP, interposing itself between the parties to a trade to guarantee settlement, and from a very early stage it introduced and developed a margin system to enhance the safety of trading [*29, 30, CME Group Margins]. This step proved a decisive turning point, dramatically raising confidence in the market and drawing far more participants into it.

The Great Pivot to Financial Futures: The Maverick Leo Melamed and the Birth of Currency Futures



After the Second World War, the CME continued to grow, centered on agricultural futures such as livestock and meat. In the late 1960s, however, the arrival of a young leader would dramatically rewrite the CME's destiny. His name was Leo Melamed [*30, CME Group History]. He held a firm conviction that the machinery of futures trading developed in the

agricultural world — that is, the techniques for managing future risk — was bound to prove useful in the volatile world of finance as well.

The historic turning point that vindicated his foresight arrived in 1971. The “Nixon shock” brought down the fixed exchange rate regime (the Bretton Woods system) that had long underpinned the world's monetary order, and the major currencies shifted en masse to floating rates. An era had dawned in which the value of

currencies fluctuated daily, and companies and investors faced new risks. Melamed did not let this once-in-a-lifetime opportunity slip. He pushed forcefully for the idea of “currency futures” as a groundbreaking financial tool for hedging the risk of floating exchange rates. Despite considerable initial skepticism, under his unwavering leadership the CME achieved a historic feat in 1972, becoming the first exchange in the world to list futures on major currencies.

This was far more than the launch of a new product. It was a revolutionary event in the history of finance, opening the door to the rich variety of financial futures and options markets that followed and that connect to the present day — interest rate futures (listed by the CBOT in 1975), equity index futures (listed by the CME in 1982), and many more. It was the moment the CME underwent a dramatic transformation, from a Chicago agricultural exchange into a global risk management center to which financial risks from around the world converge.

Innovation in Risk Management: The Impact of the SPAN® Margining System

As the new market for financial derivatives expanded rapidly, ever more sophisticated risk management techniques came to be required. In particular, a major challenge had emerged: how to measure accurately the risk of the complex portfolios in which investors combined multiple different types of futures and options, and how to calculate margin requirements commensurate with that risk. In response to this challenge — and drawing also on the lessons of the global stock market crash of 1987 (Black Monday) — the CME brought a groundbreaking margin calculation system into the world in 1988: SPAN® (Standard Portfolio Analysis of Risk) [*32, CME Group Margins]. What made SPAN® revolutionary was that, rather than calculating margin separately for each individual product, it evaluated the risk of an investor’s positions as a whole — the portfolio — using computer simulation. By positing a range of market scenarios and computing the worst loss that could

plausibly occur, it derived the required margin amount. Under this approach, the effects of risks offsetting one another across different positions were appropriately recognized, allowing investors to manage risk efficiently with less capital. SPAN®'s effectiveness won recognition around the world almost overnight; it was adopted by major exchanges and CCPs everywhere, becoming the de facto global standard and building an unshakable foundation for risk management in the modern derivatives markets.

The CME's margin regime, it should be noted, rests on two pillars: initial margin (IM), posted when a position is opened, and variation margin (VM), which settles the gains and losses arising from daily movements in market prices. Through these two mechanisms, the risks of market participants are rigorously managed day by day.

Strength in Crisis: Weathering the 2008 Global Financial Crisis

A CCP's risk management capabilities prove their true worth when markets come under unimaginable stress. The failure of Lehman Brothers in 2008 was an unprecedented crisis that shook the global financial system to its foundations, yet in that historic ordeal CME Clearing gave a magnificent demonstration of its exceptional resilience.

Lehman Brothers was one of the CME's major clearing members and, at the time of its failure, held an enormous volume of derivatives positions. Had these not been handled properly, the consequences for the market could have been incalculable. But CME Clearing acted calmly and swiftly. Following its pre-established default management procedures, it drew on the margin Lehman had posted and on the default fund resources that had been set aside for precisely such a contingency. By transferring Lehman's positions to other sound clearing members and putting them up for auction, it succeeded in resolving them in a remarkably orderly fashion within a comparatively short period, while keeping the impact on the market to a minimum.

The role CME Clearing played in preventing the kind of serious chain of settlement failures seen in the OTC (bilateral) markets, and in stopping the financial crisis from spreading further — systemic risk — was immeasurable. The episode became a lesson that demonstrated to the world anew how vital a CCP with a robust risk management framework and an ample financial foundation can be.

Relentless Evolution: The Quest for the Future Continues

The story of CME Group is one of continuous innovation — never resting on past glories, always responding to changing markets and customer needs.

- **Diversification of products and services:** The CME continually develops and brings to market new products that answer the demands of the times: futures and options on crypto assets (Bitcoin and Ether), micro products that can be traded in smaller sizes, SOFR futures referencing the new benchmark interest rate, and derivatives linked to environmental, social, and governance (ESG) themes, among others.
- **Expansion of the clearing franchise:** Beyond traditional futures and options, the CME also offers clearing services for OTC interest rate swaps and credit derivatives, which are traded largely between banks. More recently, it has been pushing actively into the clearing of cash markets at the very core of the financial system — U.S. Treasuries and repo transactions — seeking to contribute to greater efficiency and risk reduction across financial markets as a whole.
- **Harnessing technology:** The CME continually upgrades the performance of Globex, its high-speed, high-capacity electronic trading platform, provides APIs that allow customers' systems to connect with ease, and applies data analytics to enhance market transparency.
- **Pursuing new models:** Anticipating changes in market structure, the CME is also actively introducing and testing new arrangements that allow customers to deploy their funds more efficiently — for example, direct funding models under which

margin is deposited directly with the CCP and can be held on an interest-bearing basis.

Conclusion: A Clearing Giant with Innovation in Its DNA

From its beginnings as a small agricultural exchange in Chicago, the CME created the financial futures markets through the foresight of an exceptional leader, Leo Melamed; set the global standard with SPAN®, a groundbreaking risk management system; and served as a shock absorber safeguarding market stability during the financial crisis. Its journey is nothing less than the history of the development of the derivatives markets and the clearing system itself. If the Dojima Rice Exchange of the Edo period and the London Stock Exchange of the nineteenth century each laid the foundations of the clearing function for its own era, the CME can be said to have harnessed the full power of financial engineering and technology to raise that machinery to an incomparably higher level of sophistication, establishing the global standard of the modern age.

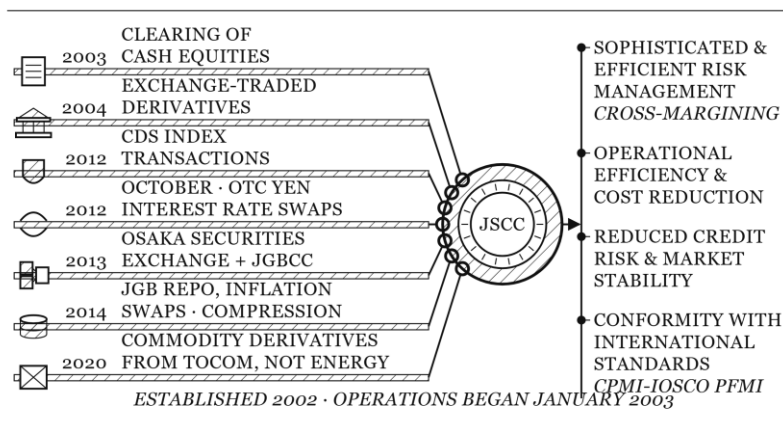
The CME Group we see today — developing one new product after another, broadening the scope of clearing, and continually taking on new technologies and models with an eye to the future — speaks eloquently to how clearing is no static piece of infrastructure, but a dynamic, vital institution that evolves in step with changing markets. That unflagging passion for innovation is the greatest driving force behind the CME's rise to its unassailable position today, and behind its shaping of the financial markets of the future.

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Column D: The Core Infrastructure Underpinning Japan's Capital Markets: The Evolution and Challenges of JSCC



Introduction: A Clearing DNA Handed Down from Dojima

Trace the rich history of Japan's exchanges and you arrive at the Dojima Rice Exchange of the eighteenth century, which pioneered organized futures trading ahead of the rest of the world. The wisdom of “clearing” born there — managing the risk of a counterparty's failure and safeguarding confidence in the market as a whole — has been handed down through the generations, changing form across the eras, to present-day Japan. And today, the institution that single-handedly bears the weighty responsibility for the safety and efficiency of trading in Japan's capital markets is the Japan Securities Clearing Corporation (JSCC). This column traces how JSCC came into being and how it has developed. It also sheds light on initiatives distinctive to the Japanese market, on JSCC's role as one of Asia's leading CCPs, and on the challenges it is taking on for the future.

The Road to the Birth of JSCC: Merging Divided Streams into One

For many years after the Second World War, Japan's cash markets for equities and other securities (centered on the Tokyo Stock Exchange) and its derivatives markets for futures, options, and similar instruments (centered on the Osaka Securities Exchange, the Tokyo Financial Exchange, and the commodity exchanges) each followed their own separate paths. Clearing functions — the settlement of trades and the management of their risks — were correspondingly fragmented, existing separately at each exchange and for each type of market.

This “siloed” structure, however, carried several drawbacks. For the financial institutions participating in these markets, having to post margin and collateral separately for each market was inefficient. Risk management was likewise fragmented market by market, and investment in systems tended to be duplicated. By the early 2000s, strengthening the international competitiveness of Japan's financial markets and building safer, more user-friendly market infrastructure had become an urgent imperative. To overcome this challenge, a framework was conceived that would consolidate the divided clearing functions into a single entity covering the market as a whole. The fruit of that effort was JSCC, established in 2002, which commenced operations in January 2003. This was a genuinely landmark event in the history of Japan's capital market infrastructure.

JSCC's Consolidation and Expansion of Clearing Functions: The Path to One-Stop Clearing

Since its establishment, JSCC has steadily broadened the scope of its clearing services, growing into a comprehensive clearing house that covers most of Japan's major financial and commodity markets.

- 2003:** JSCC began with the clearing of cash equities.
- 2004:** Commenced clearing of exchange-traded derivatives, such as the equity index futures and options traded on the Tokyo Stock Exchange (as it then was).

- 2012:** Commenced clearing of credit default swap (CDS) index transactions.
- October 2012:** Commenced clearing of over-the-counter (OTC) yen-denominated interest rate swaps. This was a critically important step, taken in response to the international OTC derivatives reforms agreed at the G20 summits in the wake of the 2008 global financial crisis.
- 2013:** Integrated the derivatives clearing functions of the Osaka Securities Exchange (as it then was) and the functions of the Japan Government Bond Clearing Corporation (JGBCC), which had specialized in clearing Japanese government bonds. This established a framework under which the major exchange-traded derivatives and government bonds are cleared centrally within JSCC.
- 2014:** Commenced clearing of JGB repo transactions and inflation swaps (OTC derivatives). In December of that year, JSCC also launched a compression service for interest rate swaps, which reduces risk by tearing up economically redundant contracts [^{*33}, JSCC News 2014/12/05].
- 2020:** Took over the commodity derivatives clearing functions — covering precious metals, rubber, agricultural products, and more — from the Tokyo Commodity Exchange (TOCOM), thereby commencing the clearing of the major commodity derivatives other than energy products.

Through this series of integrations and expansions, JSCC has evolved into one of the world's leading CCPs, offering a one-stop clearing service that handles an exceptionally diverse range of asset classes — equities, bonds, exchange-traded derivatives, OTC derivatives (interest rate swaps and CDS), and commodity derivatives — within a single clearing house.

Confront

The consolidation of clearing functions under JSCC has brought many benefits to Japan's markets.

- **More sophisticated and more efficient risk management:** The risk of positions spanning multiple markets can now be captured and managed on a consolidated basis, improving the precision of risk management. Moreover, by unifying margin calculation across different asset types (through cross-margining and similar arrangements), JSCC reduces the margin burden on market participants and allows them to deploy their capital more efficiently.
- **Operational efficiency and cost reduction:** The standardization of clearing systems and procedures has also reduced the administrative burden and costs borne by market participants as a whole.
- **Reduced credit risk and market stability:** By interposing itself as the CCP in every trade, JSCC severs the credit risk between individual participants. As a result, even if a particular financial institution fails, the systemic risk that its failure will cascade through the entire market can be effectively contained.
- **Conformity with international standards:** By maintaining rules and operations that conform fully to the international standards set by CPMI-IOSCO (the Committee on Payments and Market Infrastructures and the International Organization of Securities Commissions) — the Principles for Financial Market Infrastructures (PFMI) — JSCC offers a market environment that overseas investors can trust and trade in with confidence.

At the same time, the potential challenges of concentrating clearing functions in a single institution are also recognized. These include the magnitude of the impact should a large-scale failure occur in JSCC's systems, and the inefficiencies that can arise as an organization grows larger. JSCC continues to address these challenges in

earnest, through measures such as duplicating and triplicating its systems, strengthening its backup arrangements (business continuity planning, BCP), and continuously reviewing how the organization is run.

JSCC's Risk Management: World-Class Reliability

JSCC has built — and operates day in and day out — a sophisticated risk management framework in line with international standards.

- **A rigorous margin regime:** JSCC rigorously collects from its clearing participants initial margin (IM), posted when a trade is initiated to provide for potential future losses, and variation margin (VM), which covers the gains and losses generated by daily market price movements (mark-to-market). Initial margin is calculated using tools such as value-at-risk (VaR) models, which statistically analyze market risk [^{*34}, JSCC Website].
- **A multi-layered response to default (the default waterfall):** Against the contingency that a clearing participant defaults, JSCC maintains multiple layers of resources to absorb the resulting losses. Specifically, it has established a default waterfall under which losses are absorbed in the following order: (i) first, the margin and default fund contributions posted by the defaulting participant itself; (ii) next, resources contributed by JSCC itself; and (iii) if these prove insufficient, the default fund contributions of the other, non-defaulting participants.
- **Stress testing and fire drills:** JSCC regularly runs tests based on market movements far more severe than anything seen under normal conditions (stress scenarios), confirming that it can secure sufficient financial resources under any circumstances. It also works continuously to enhance its crisis-response capabilities, including by joining major overseas CCPs in joint simulation exercises (fire drills) that assume an actual default has occurred.
- **Conformity with international standards:** By maintaining rules and operations

that conform fully to the international standards set by CPMI-IOSCO (the Committee on Payments and Market Infrastructures and the International Organization of Securities Commissions) — the Principles for Financial Market Infrastructures (PFMI) — JSCC has obtained licenses to operate as a clearing house from the authorities of seven overseas jurisdictions, offering a market environment that overseas investors can trust and trade in with confidence.

Through these steady, painstaking efforts, JSCC has earned a strong international reputation and established itself firmly as one of the leading CCPs in the Asian region.

Embracing Innovation and JSCC's Role in Asia

Beyond operating its day-to-day business safely and reliably, JSCC is also actively pursuing innovation to meet the evolving needs of the market. It continues to advance on every front: expanding the range of products it can clear (for example, single-name CDS clearing, which covers the credit risk of individual companies); broadening the scope of margin offsets across different asset classes (cross-margining); making collateral management more efficient (for example, by making foreign currencies easier to use as collateral); and exploring the potential use of new technologies, such as distributed ledger technology (DLT), in the future.

JSCC also contributes to the stability and development of financial markets across the Asian region as a whole — by deepening its cooperation with other CCPs in Asia, exchanging information, and contributing to discussions on international rule-making.

Conclusion: A Clearing House Carrying Japan's Trust into the Future

Japan's long history of trading and clearing, which began with the Dojima Rice Exchange, lives on today in the birth and development of JSCC as a modern, comprehensive CCP. By unifying what had been fragmented clearing functions, JSCC dramatically improved both the efficiency and the safety of Japan's capital markets. And by building a robust risk management framework that conforms to international standards, it has earned the unwavering trust of market participants at home and abroad.

While squarely confronting challenges such as the centralized management of system risk and the governance of a larger organization, JSCC has rolled out one innovative service after another — cross-margining, compression, and more — and is raising its profile as one of Asia's flagship CCPs. JSCC's work forms a vital foundation for the further development of Japan's capital markets amid global competition, and it is no exaggeration to say that its evolution and ambitions will shape the very future of Japan's financial infrastructure. Building on the history and experience it has accumulated, JSCC is widely expected to keep contributing to the stability and development of the market.

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Column E: Does History Repeat Itself? Lessons from Derivatives Loss Events and the Evolution of Clearing

Introduction: Learning from the History of Derivatives and Crises

Derivatives are indispensable tools in modern financial markets for managing risk and executing investment strategies. Yet a look back at their history reveals another truth: at times they harbor a terrifying destructive power, capable of shaking the entire financial system and driving even venerable firms with centuries of history into failure in an instant. From the frenzy of the tulip mania in seventeenth-century Holland, through the collapse of Barings Bank in 1995, the LTCM crisis of 1998, and the 2008 global financial crisis, to the still-fresh memories of the Archegos shock of 2021, the FTX collapse of 2022, and the rescue merger of Credit Suisse in 2023, there has been no end to the loss events and market disruptions in which derivatives trading was a direct or indirect trigger.

These painful experiences have taught us, time and again, how serious the risks latent in derivatives trading are and how difficult they are to manage. Rather than dismissing past failures as mere incidents and letting them fade from memory, we must dig deep into why they occurred, identify the structural problems, and draw concrete lessons — and then apply those lessons to the design of future risk management practices and institutional frameworks. That is the essential responsibility falling on all of us who seek to build a stronger and more resilient financial system. This column looks back at the major derivatives-related loss events in history and analyzes the problems that lay behind them. It then considers how these bitter experiences have driven the evolution of clearing as an institution — and, in particular, of the role of the central counterparty (CCP).

Past Derivatives Loss Events: Causes and Lessons

The Collapse of Barings Bank (1995)

● **Overview:** Barings Bank, a prestigious British merchant bank with a glorious history of more than 200 years, suffered an astronomical loss of approximately GBP 1.3 billion (roughly JPY 230 billion at the exchange rate of the time) as a result of unauthorized trading in Nikkei 225 futures and options by Nick Leeson, a trader at its Singapore office, and was driven into failure [Rawnsley 1995].

● **Causes:** Leeson initially took ever-larger positions in an attempt to recover what were at first relatively small losses. As the losses snowballed, he concealed them by booking fictitious trades and abusing a special

internal account (the infamous “Error Account 88888”). The root causes of the affair were: (i) a

fatal flaw in internal controls, in that the trader

himself was responsible both for executing trades

and for the subsequent back-office processing

(settlement and record-keeping); (ii) a risk

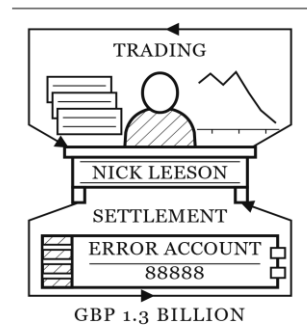
management function that failed to operate

effectively, with **ambiguous reporting lines** to

senior management; and (iii) the fact that **London**

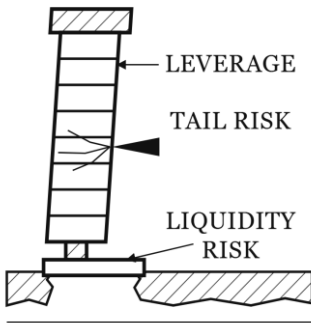
headquarters did not exercise adequate oversight of its rapidly growing Asian operations.

● **Lessons:** The affair threw into sharp relief the importance of enforcing segregation of duties — the bedrock of internal control; of establishing a risk management function independent of the traders and endowing it with sufficient authority; and of senior management itself understanding the importance of risk management and building an effective supervisory framework.



The LTCM Crisis (1998)

- **Overview:** Long-Term Capital Management (LTCM) was a hedge fund run by a veritable “dream team” that included two Nobel laureates in economics. Its



strategy of exploiting price differentials between bonds using sophisticated mathematical models (arbitrage) generated enormous profits for a time. But the Russian financial crisis of the summer of 1998, and the global market turmoil that followed (a flight to safe assets), inflicted losses of approximately USD 4.6 billion within a very short period, bringing the fund to the brink of collapse. In the end, the Federal Reserve (the U.S. central bank),

concerned about the repercussions for the financial system as a whole, took the exceptional step of brokering a rescue carried out by a group of private financial institutions [Lowenstein 2000].

- **Causes:** Several factors intertwined: (i) **extremely high leverage** (investing with borrowed funds), amounting to many dozens of times the fund’s capital; (ii) **models that leaned too heavily on historical data and proved fragile** in the face of extreme market moves (tail risk) that were “never supposed to happen” (the assumption that correlations would not change); (iii) a **disregard for liquidity risk** — the risk that, when markets are disrupted, positions cannot be sold even when one wants to; and (iv) the fact that many financial institutions had similar trades on to LTCM’s, so that **the market had become prone to tipping in one direction** (crowding risk).
- **Lessons:** The crisis taught several important lessons: that no financial engineering model, however sophisticated, is infallible, and that one must always remain conscious of **the risks inherent in the models themselves** and of the possibility

of unforeseen events (tail risk); the importance of managing leverage appropriately and conducting stress tests based on severe scenarios; the need for risk management that allows for the possibility of liquidity draining from the market; and vigilance to how correlations between apparently unrelated markets **can rise unexpectedly in times of crisis**.

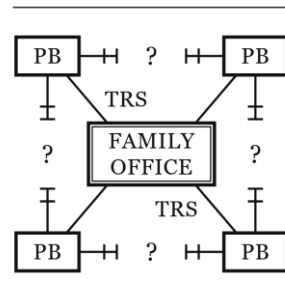
The Global Financial Crisis (2008)

- **Overview:** What began as a problem with U.S. subprime mortgages — housing loans to borrowers with weak credit — escalated into a global financial crisis when the prices of the complex financial products into which those loans had been packaged (securitized products such as MBS and CDOs) collapsed. In September 2008, Lehman Brothers, a prestigious U.S. investment bank, failed under the weight of massive losses and write-downs related to these securitized products (with total liabilities of approximately USD 613 billion) [FSA 2010]. This historic failure dramatically amplified mutual distrust among financial institutions, paralyzing the markets for short-term funding and triggering a worldwide collapse in equity prices.
- **Causes:** Many factors can be cited: excessive subprime lending and the accommodative monetary policy that encouraged it; the mass origination of complex securitized products whose risks were exceedingly difficult to assess; overreliance on the assessments of rating agencies; and lax risk management (excessive risk-taking) on the part of financial institutions themselves. What deepened the crisis above all, however, was the **chain reaction of counterparty risk** running through the lightly regulated and **opaque over-the-counter (OTC) derivatives market** (especially CDS). The pervasive distrust — no one knew who was carrying how much risk — plunged the entire market into panic.
- **Lessons:** Strengthening the risk assessment and risk management of complex

financial products; rethinking excessive reliance on credit ratings; and, above all, **improving the transparency of the OTC derivatives market** (mechanisms for aggregating and reporting trade data), **strengthening counterparty risk management** (mandatory central clearing through CCPs and the introduction of margin requirements), and **containing risk to the financial system as a whole (systemic risk)** — these came to be recognized as the highest priorities for the international community, leading to the sweeping regulatory reforms based on the agreements reached at the G20 summits.

The Archegos Shock (2021)

- **Overview:** Archegos Capital Management was a family office (a firm managing personal wealth) run by Bill Hwang. Acting through several major securities firms (prime brokers, PBs), it made extremely highly leveraged, concentrated investments in particular stocks, mainly through equity derivatives known as total return swaps (TRS). When the share prices of its investments plunged, however, it became unable to meet additional margin calls and effectively collapsed. As a result, the major global financial institutions on the other side of its trades — including Nomura Securities and Credit Suisse — suffered combined losses exceeding USD 10 billion. [Kolhatkar 2022]



- **Causes:** Several factors operated in combination: (i) the use of the family office structure, which is more lightly regulated than hedge funds and subject to only limited disclosure obligations; (ii) the fact that, because Archegos traded through multiple PBs, **no individual PB could grasp the full picture of its risk (its total exposure)**; (iii) **extreme leverage** obtained through the TRS (trading at

many times the scale of its own capital); (iv) **excessive concentration** in particular stocks; and (v) **lax risk management on the part of the PBs** that were its counterparties (excessive extension of credit, deficient collateral valuation, and so on).

- **Lessons:** The episode forcefully renewed several questions: the need to strengthen **regulation and oversight of private funds**, including family offices; the need for PBs to **fundamentally rethink their management of client risk (especially leverage and concentration risk)**; the need to **improve the transparency of derivatives transactions** such as equity swaps (for example, by considering position-reporting obligations); and the importance of a sound **risk culture** across financial institutions as a whole.

The Role of Clearing: A Mechanism for Mitigating Risk and Its Evolution

These historic loss events have shown us, in the starkest terms, how destabilizing to the entire market the risks accompanying derivatives trading can be — above all counterparty risk and informational opacity. Clearing is the critical piece of market infrastructure for mitigating these risks, and the central role in clearing is played by the CCP.

Just as the *chōai* book-offsetting practice of Japan's Dojima Rice Exchange made it possible to settle trades without identifying one's counterparty, the basic idea of clearing has existed since ancient times. The modern CCP can be seen as a vastly more sophisticated, systematized embodiment of that idea.

- **Absorption of counterparty risk:** The CCP becomes the buyer to every seller and the seller to every buyer, interposing itself as the counterparty to each trade (**assumption of obligations / novation**). The credit risk that the original trading parties bore directly toward each other is thereby severed, and each participant comes to contract solely with the highly creditworthy CCP rather than with individual counterparties.

- Risk protection through margin:** The CCP collects initial margin (IM) and variation margin (VM) from its participants. IM is collateral held against the potential losses that future market movements could generate (potential future exposure), and is calculated according to the riskiness of the position. VM covers the valuation gains and losses arising from daily market price movements (mark-to-market) and is, as a rule, exchanged every day. This prevents losses from ballooning without limit. (The minimum margin level required to maintain a position is known as maintenance margin; when the value of the initial margin account falls below this level, additional margin is demanded — a margin call.)
- Guarantee of settlement:** On the basis of rigorous risk management and ample financial resources held against the contingency of a failure (the multi-layered loss-absorbing buffers known as the default waterfall), the CCP guarantees the performance of settlement even if a participant defaults.
- Greater transparency:** Because information about trades is concentrated at the CCP, regulators and the market as a whole can more readily grasp the state of risk.

The Role of CCPs in the Global Financial Crisis: A Shock Absorber Against the Crisis

In the 2008 global financial crisis, the role played by CCPs — notably LCH SwapClear, which cleared interest rate swaps, and CME Clearing, which cleared a wide range of derivatives — was decisively important in preventing the crisis from spreading further. On the bankruptcy of Lehman Brothers, these CCPs swiftly activated their pre-established default management procedures. Drawing on the enormous margin Lehman had posted, and while keeping the impact on the market to a minimum, they succeeded in processing Lehman's vast derivatives positions — transferring them to other sound clearing members or putting them up for auction — within a relatively short period and in a remarkably orderly fashion.

Had these trades not been cleared through CCPs, it is widely held that the chain reaction of counterparty risk would have grown even more severe, and the collapse of the financial system could have been far more devastating. This experience left a powerful impression around the world of just how effective a risk mitigation tool the CCP is, and it became a decisive factor propelling the subsequent G20 reforms of the OTC derivatives market (**mandatory central clearing and strengthened margin requirements**).

Conclusion: Carrying the Lessons Forward

Barings, LTCM, Lehman, Archegos — these episodes unfolded in different eras, in different markets, and involved different products, yet beneath them all the same factors recur: **excessive leverage, flawed risk management, deficient internal controls, and a lack of transparency**. History tends to repeat itself, even as it changes form. If derivatives — tools that can be extraordinarily powerful in the right hands — are to be used safely, we must learn from these lessons with humility and work continuously to review and strengthen our risk management frameworks.

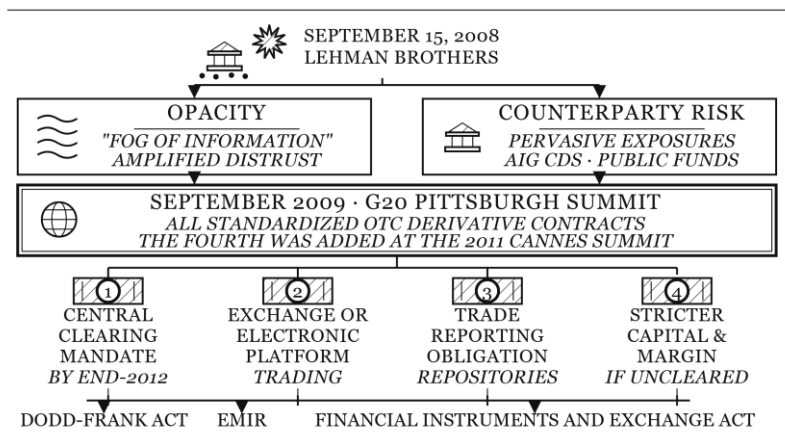
Clearing — and central clearing through CCPs in particular — has proven to be a highly effective means of dramatically reducing counterparty risk and enhancing market transparency. Yet CCPs are no panacea: they give rise to new risks of their own, including concentration risk and operational risk. The key to financial stability in the years ahead lies in three things working together: the international effort to maintain and strengthen the resilience of CCPs themselves (compliance with the PFMI, rigorous stress testing, and the development of recovery and resolution plans); the soundness of the clearing members (clearing brokers) that support them; and, ultimately, the risk discipline of every individual market participant.

The world of derivatives and clearing will not stop evolving. New products will appear, new technologies will emerge, and, in all likelihood, new kinds of risk will arise as well. To keep the universal lessons of past loss events firmly in mind and to keep evolving our institutions and practices in a spirit of transparency, sound risk management, and international cooperation — that is the only path to preventing future crises and achieving the sustainable development of our markets.

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Column F: The Bankruptcy of Lehman Brothers and the G20 Agreement:



The Path to Transforming OTC Derivatives Clearing

The Fragility of the OTC Derivatives Market Exposed by the Global Financial Crisis

On September 15, 2008, Lehman Brothers, one of America's most prestigious investment banks, failed without warning. This shocking event shook financial markets around the world to their foundations and marked the opening of a crisis that would be etched into history — the 2008 global financial crisis. Triggered by the subprime mortgage problem, the crisis laid bare just how fragile the bewilderingly complex securitized products had become, and how fragile the over-the-counter (OTC) derivatives market that had served to transfer and disperse their risks throughout the financial system had become [*FSA 2010].

At the time, the OTC derivatives market — then the dominant venue — consisted of financial institutions trading directly with one another on a bilateral basis, without

going through any public venue such as an exchange. As a result, it suffered from serious structural weaknesses, including the following.

- **Opacity:** Who was trading what, and on what scale? For market participants and regulators alike, gauging the true state of the market was extremely difficult. After the failure of Lehman Brothers, this “fog of information” amplified mutual distrust among financial institutions and became a major driver of the panic-driven credit contraction in which no one was willing to lend.
- **Pervasive Counterparty Risk:** The risk that a counterparty would fail to perform as promised (counterparty risk) ran through the entire market like an intricate web. When a single institution as large as Lehman failed, losses cascaded to the many financial institutions that had been its counterparties, making real the specter of systemic risk — the danger that the entire financial system could seize up. The case of AIG, the largest insurer, which accumulated enormous losses on its credit default swap (CDS) positions and had to be rescued with public funds, was emblematic of just how vast and dangerous counterparty risk in the OTC market had become.

At an exceedingly high price, the global financial crisis drove home to the world how profoundly these structural flaws in the OTC derivatives market could threaten the stability of the entire financial system.

The G20 Pittsburgh Summit: The Starting Gun for Internationally Coordinated Reform

The international community responded to this unprecedented crisis with a speed and unity never seen before. In September 2009, roughly one year after the bankruptcy of Lehman Brothers, the G20 (Group of Twenty) summit held in Pittsburgh reached a historic agreement on reforms that would fundamentally reshape the OTC derivatives market. Its core commitments can be distilled into the

following three points [*G20 Leaders Statement 2009].

- 1. Central Clearing Mandate:** All standardized OTC derivative contracts should be cleared through central counterparties (CCPs) by end-2012 at the latest.
- 2. Exchange or Electronic Platform Trading:** Standardized OTC derivative contracts should be traded on exchanges or electronic trading platforms, where appropriate.
- 3. Trade Reporting Obligation:** Information on all OTC derivative contracts must be reported to trade repositories.

In addition, at the 2011 Cannes Summit it was agreed that OTC derivative transactions not cleared through CCPs would be subject to more stringent capital and margin requirements.

The G20 Pittsburgh agreement charted a clear course for bringing discipline to the OTC derivatives market and reducing its risks. It went on to serve as the unshakable international foundation on which concrete regulations were built in each jurisdiction — the Dodd-Frank Act in the United States, EMIR in Europe, and the amendments to the Financial Instruments and Exchange Act in Japan, among others. And at the very heart of the reforms stood none other than mandatory central clearing through CCPs.

The Role of CCPs During the Global Financial Crisis and the Impetus for Reform

Why did the G20 place central clearing through CCPs at the heart of its reforms? The reason lay in a simple fact: under conditions that amounted to an extreme stress test — the global financial crisis — several CCPs already in existence had convincingly demonstrated their risk management capabilities and their effectiveness. In particular, the SwapClear service of LCH.Clearnet Ltd, which cleared the bulk of interest rate swaps, and CME Clearing, which handled a wide range of derivatives,

managed to wind down the enormous derivatives positions of the defaulted Lehman Brothers relatively smoothly, without triggering major disruption in the markets

[*Davies 2017, CCP12 2017].

When Lehman defaulted, these CCPs covered the resulting losses by drawing, in accordance with pre-established procedures, on the multiple layers of the default waterfall: first, the margin Lehman itself had posted; second, the funds Lehman had contributed to the CCP's mutualized default fund; and third, where those proved insufficient, the contributions of the surviving clearing members and the CCP's own capital. At the same time, they executed the default management process swiftly and precisely — porting Lehman's positions to other, sound clearing members, putting on hedges to neutralize market risk, and auctioning the residual positions [*Davies 2017]. It was precisely this performance by the CCPs that contained the transmission of counterparty risk along the chain to other financial institutions; they played a decisive role in damping market-wide panic [*Tucker 2018]. This track record gave powerful impetus to the international argument that clearing through CCPs should be extended across the entire OTC market.

The Road to Mandatory Central Clearing: Challenges to Overcome

Moving from the agreement reached at the G20 summit to the actual introduction of mandatory clearing rules in each jurisdiction required a complex process: the development of international standards (led by the Financial Stability Board (FSB), the International Organization of Securities Commissions (IOSCO), and the Basel Committee on Banking Supervision (BCBS), among others), followed by legislation in each country built on those standards.

Along the way, a number of important issues were vigorously debated.

●**Scope of the Clearing Mandate:** Which types of derivatives should fall within the mandate? Initially, the focus fell on interest rate swaps and CDS, which traded in

large volumes and whose contract terms were reasonably standardized.

- **Requirements for CCPs:** What standards should a CCP meet to shoulder the heavy responsibility of central clearing? Detailed international standards covering risk management frameworks, governance, financial resources, and more were established (the CPMI-IOSCO Principles for Financial Market Infrastructures, the PFMI) [^{*}CPMI-IOSCO 2012].
- **Cross-Border Application:** To what extent should one jurisdiction's rules apply to counterparties located abroad, or to transactions executed abroad? This proved a particularly contentious issue, with regulatory approaches diverging across jurisdictions. The U.S. CFTC, for example, pursued a relatively broad extraterritorial reach, whereas Japan initially focused on transactions booked domestically. Such differences also gave rise to concerns about market fragmentation and about regulatory arbitrage — the migration of trading to jurisdictions with lighter regulation.
- **Margin Requirements for Non-Cleared Trades:** Rules strictly requiring the exchange of initial margin and variation margin for trades not cleared through a CCP [BCBS-IOSCO 2019 Margin Requirements]. While these rules created an incentive to clear through CCPs, they also increased the collateral burden on market participants.
- **Costs and the Impact on Market Liquidity:** Using a CCP entails fees and margin obligations. Industry participants voiced concerns that mandatory central clearing and the associated tightening of regulation would raise costs for market participants and, as a result, impede trading and reduce liquidity. Fee increases did in fact occur, as at LCH. On these complex issues, regulators in each jurisdiction developed and introduced the rules in stages, in continual dialogue with market participants.

The Significance of Mandatory Central Clearing and the Challenges That Remain

Building on the lessons of the global financial crisis, the OTC derivatives market reforms pursued under the G20 agreement — above all, mandatory central clearing through CCPs — were unquestionably a major step forward in strengthening the stability of the financial system. The counterparty risk that once blanketed the market has been reduced substantially, and market transparency has improved dramatically.

That said, not every challenge has been resolved.

- **Concentration of Risk in CCPs:** The counterparty risk once dispersed across the entire market is now concentrated in CCPs. While this makes risk management more efficient, the consequences should a CCP itself ever fail would be incalculable. Strengthening the supervision of CCPs, and putting in place frameworks for resolving a failing CCP in an orderly manner while minimizing the impact on markets (resolution), have therefore become urgent international priorities. Conducting rigorous stress tests (“fire drills”) based on severe scenarios — for individual CCPs and across multiple CCPs — is also important for verifying their resilience.
- **Cost Burden:** The fees and margin associated with clearing remain a heavy burden, particularly for small and mid-sized market participants. Healthy competition among CCPs to bring down costs, and the development of more efficient collateral management techniques, will continue to be needed.
- **Impact on Liquidity:** It has also been suggested that tighter regulation may be making certain derivative products harder to trade, thereby reducing market liquidity. Weighing the benefits of the regulations against their side effects, and fine-tuning or revisiting the rules as needed, remains a subject of ongoing debate.
- **Regulatory Complexity and International Consistency:** Detailed differences

persist among national regimes, and coping with this complex regulatory patchwork imposes a significant burden on financial institutions that trade across borders. Continued efforts toward international regulatory harmonization remain essential.

Conclusion: A Reform Journey Without End

The global financial crisis confronted the world, with shocking force, with the dangers of an OTC derivatives market into which regulators could not see clearly and in which risk had accumulated opaquely. The G20 agreement born of that bitter experience, and the series of reforms that gave it concrete form — above all, mandatory central clearing through CCPs — can fairly be said to have achieved landmark gains in strengthening the stability of the financial system. CCPs proved their effectiveness in the crucible of the Lehman crisis and, through the reforms, cemented their position as, quite literally, the core of the financial market infrastructure.

But the journey of reform does not end here. Responding to the new challenge of risk concentration in CCPs, pursuing the right balance between cost and efficiency, and adapting to ever-changing market conditions — and to new risks that may yet emerge — will continue to be demanded of us. More than a decade after the global financial crisis, we must never let the lessons of that day fade; rather, we must persevere, in a spirit of international cooperation, in our efforts to build a safer and more resilient financial system.

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About the Author

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Shun Yanagisawa is Director, Head of Futures, Clearing and FX Prime Brokerage, Markets Sales, at Citigroup Global Markets Japan Inc. In this capacity, he partners daily with institutional investors, hedge funds, and global market participants to advance market architecture — particularly across futures, options, and interest rate derivatives — while ensuring the resilient execution of post-trade infrastructure, including clearing, margin optimization, and settlement.

With a distinguished professional career spanning over two decades split between J.P. Morgan Securities Japan and Citigroup Global Markets Japan, Mr. Yanagisawa possesses rare, end-to-end expertise across the front-office and post-trade spectrum — ranging from proprietary execution and algorithmic trading to regulatory strategy and quantitative risk management. Throughout the structural expansion of OTC derivatives clearing, he has been at the vanguard of business architecture, product innovation, and client franchise development, establishing robust operating frameworks tailored to a rapidly shifting regulatory landscape.

Beyond his corporate mandate, he holds several key leadership and advisory roles within the financial industry, including serving as a Representative Director and Vice President of FIA Japan and a member of its Operations Committee, a member of the Japan Securities Clearing Corporation's (JSCC) Interest Rate Swap Steering Committee, and a member of the Asia Risk advisory board. Through sustained dialogue with global regulators, clearing houses, and market participants, he remains a vocal advocate for market practices and rules that harmonize systemic safety with capital efficiency.